



US012409203B2

(12) **United States Patent**
Jang et al.

(10) **Patent No.:** **US 12,409,203 B2**

(45) **Date of Patent:** **Sep. 9, 2025**

(54) **POLYPEPTIDE DIMER WITH HIGH SIALIC ACID CONTENT, COMPRISING EXTRACELLULAR DOMAIN OF ALPHA SUBUNIT OF IGE FC RECEPTOR, AND PHARMACEUTICAL COMPOSITION COMPRISING SAME**

(58) **Field of Classification Search**

CPC A61K 38/00; A61K 9/0019; A23L 33/18;
A61P 37/08; A61P 11/06; C07K
14/70535; C07K 2319/00; A23V
2200/304

See application file for complete search history.

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(73) Assignee: **GI INNOVATION, INC.**, Seoul (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 762 days.

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(21) Appl. No.: **17/625,668**

(22) PCT Filed: **Jul. 7, 2020**

(86) PCT No.: **PCT/KR2020/008855**

§ 371 (c)(1),

(2) Date: **Jan. 7, 2022**

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(87) PCT Pub. No.: **WO2021/006599**

PCT Pub. Date: **Jan. 14, 2021**

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(65) **Prior Publication Data**

US 2022/0257693 A1 Aug. 18, 2022

(57) **ABSTRACT**

(30) **Foreign Application Priority Data**

Jul. 8, 2019 (KR) 10-2019-0082217

(51) **Int. Cl.**

A61K 38/00 (2006.01)

A23L 33/18 (2016.01)

(Continued)

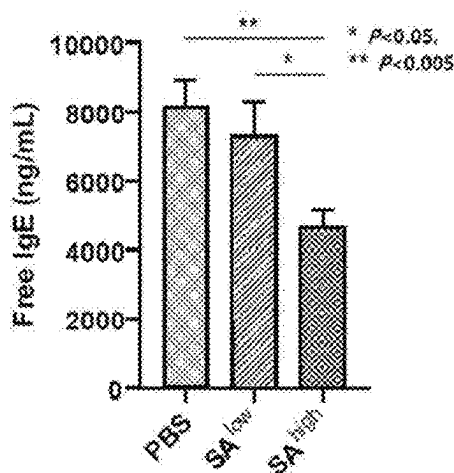
A modified IgE Fc receptor having a high content of sialic acid and a pharmaceutical composition contains the modified IgE Fc receptor are disclosed. The polypeptide dimer having a high content of sialic acid not only has excellent safety and persistence in a body as compared with conventionally used anti-IgE antibodies, but also shows strong binding to IgE. Thus, the polypeptide has the advantage of an extended administration cycle. The polypeptide dimer is also an IgE single target substance, and unlike conventional anti-IgE antibodies to which the Fc of IgG1 is applied, does not bind to an Fc gamma receptor. The polypeptide dimer

(Continued)

(52) **U.S. Cl.**

CPC **A61K 38/00** (2013.01); **A23L 33/18** (2016.08); **A61K 9/0019** (2013.01);

(Continued)



can be usefully used for the prevention or treatment of allergic diseases.

13 Claims, 18 Drawing Sheets

Specification includes a Sequence Listing.

(51) Int. Cl.

A61K 9/00 (2006.01)

A61K 9/70 (2006.01)

A61P 37/08 (2006.01)

C07K 14/735 (2006.01)

(52) U.S. Cl.

CPC *A61K 9/7023* (2013.01); *A61P 37/08* (2018.01); *C07K 14/70535* (2013.01); *C07K 2319/30* (2013.01)

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FIG.1

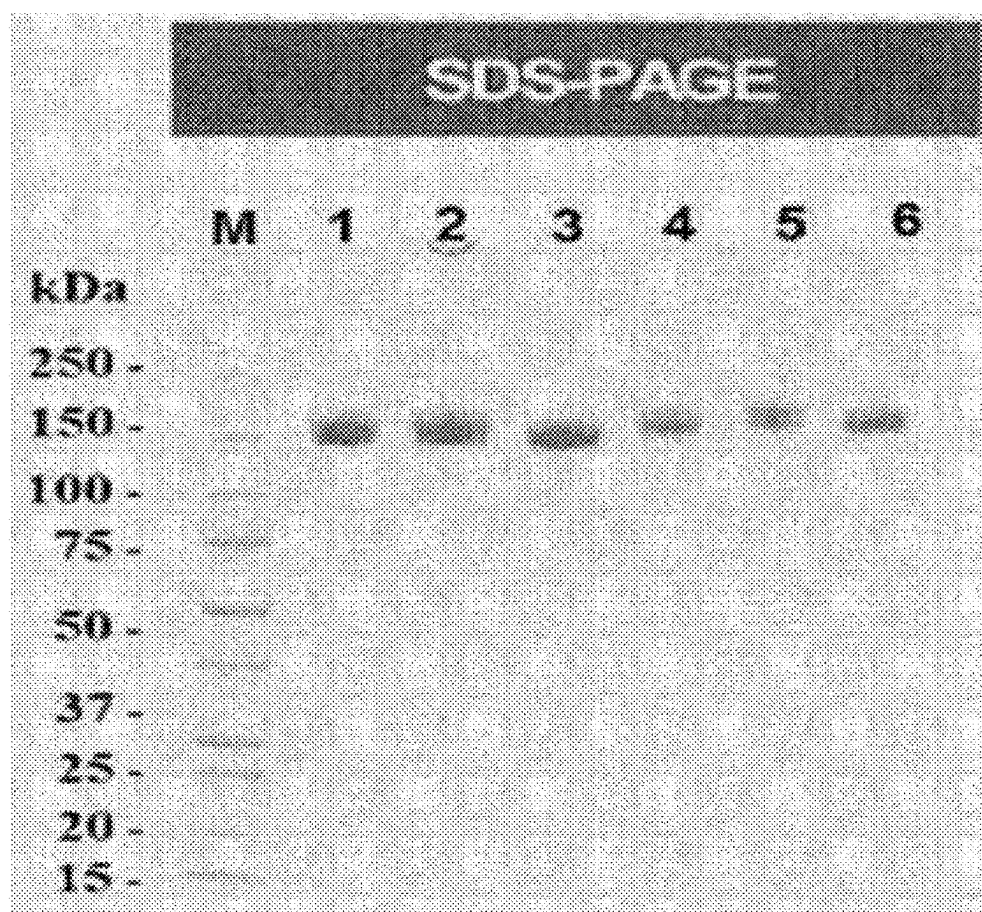
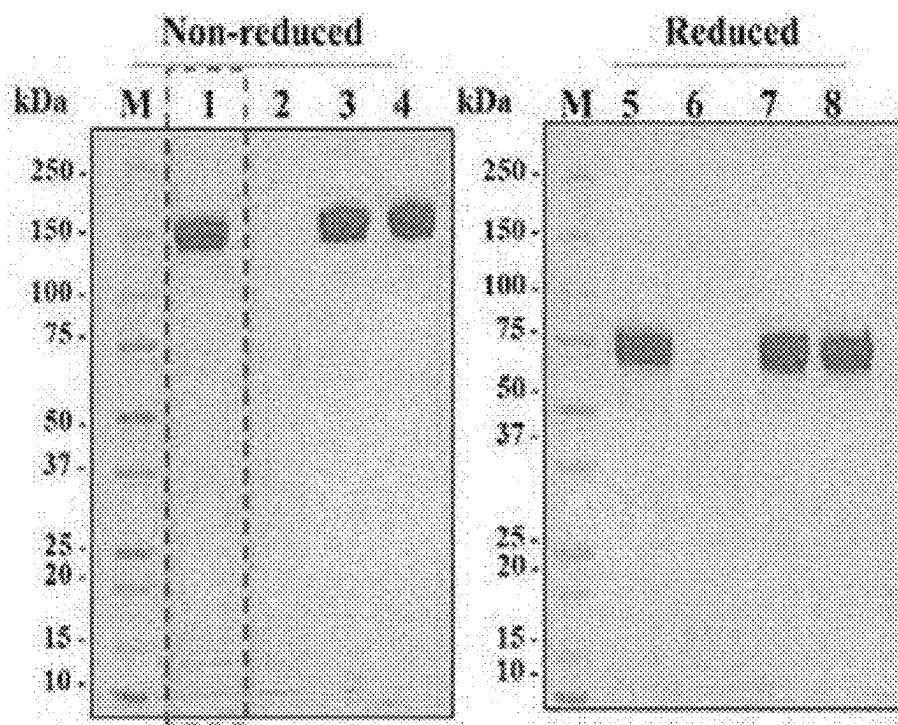


FIG.2



Lane	Sample name	Loading amount
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2	FT	Max
3	Elute	3 ug
4	Elute_N	3 ug
5	Input	Max
6	FT	Max
7	Elute	3 ug
8	Elute_N	3 ug

FIG.3

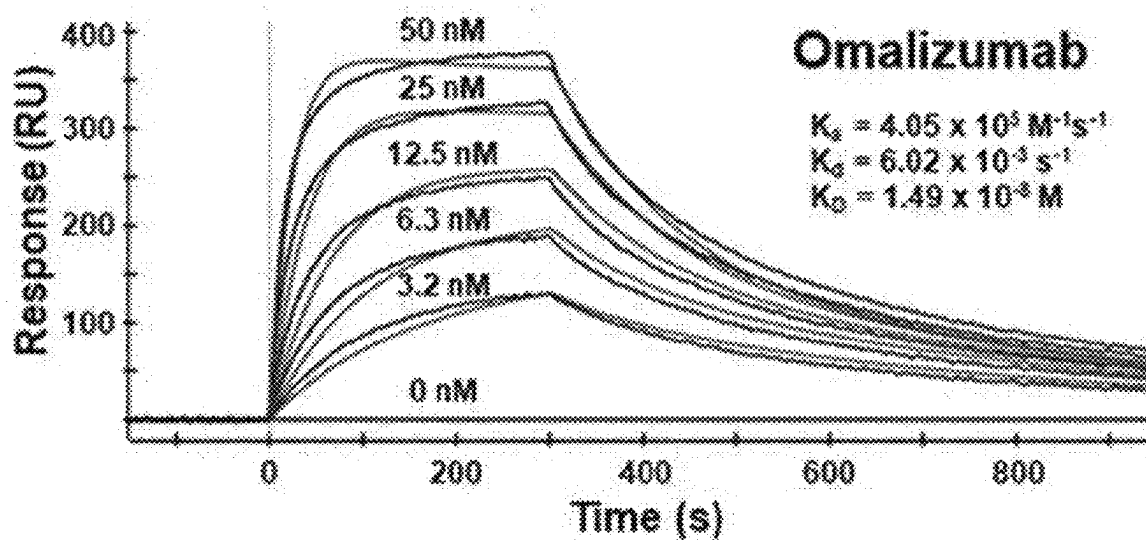


FIG.4

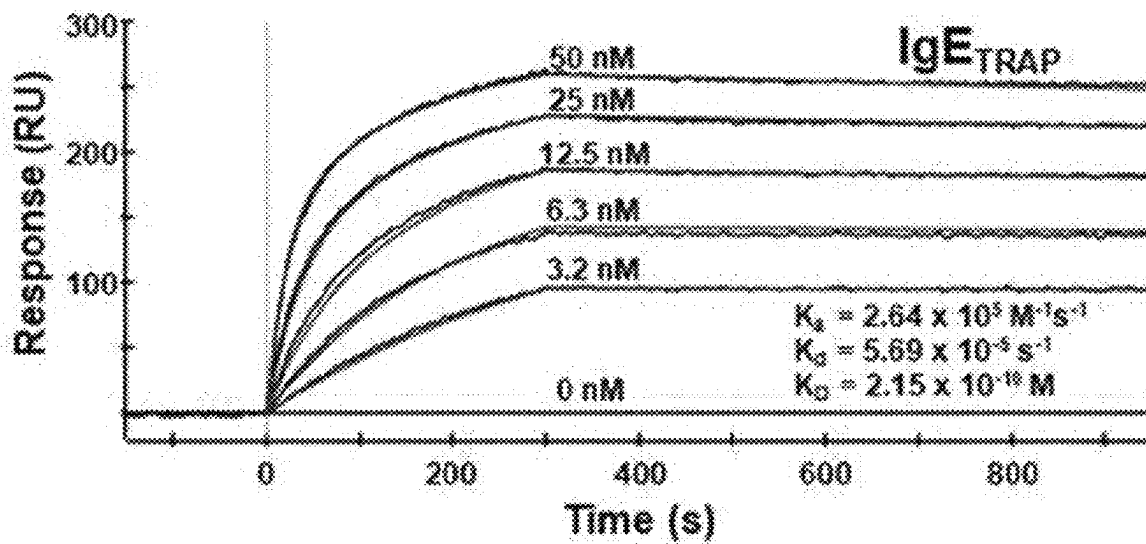


FIG.5a

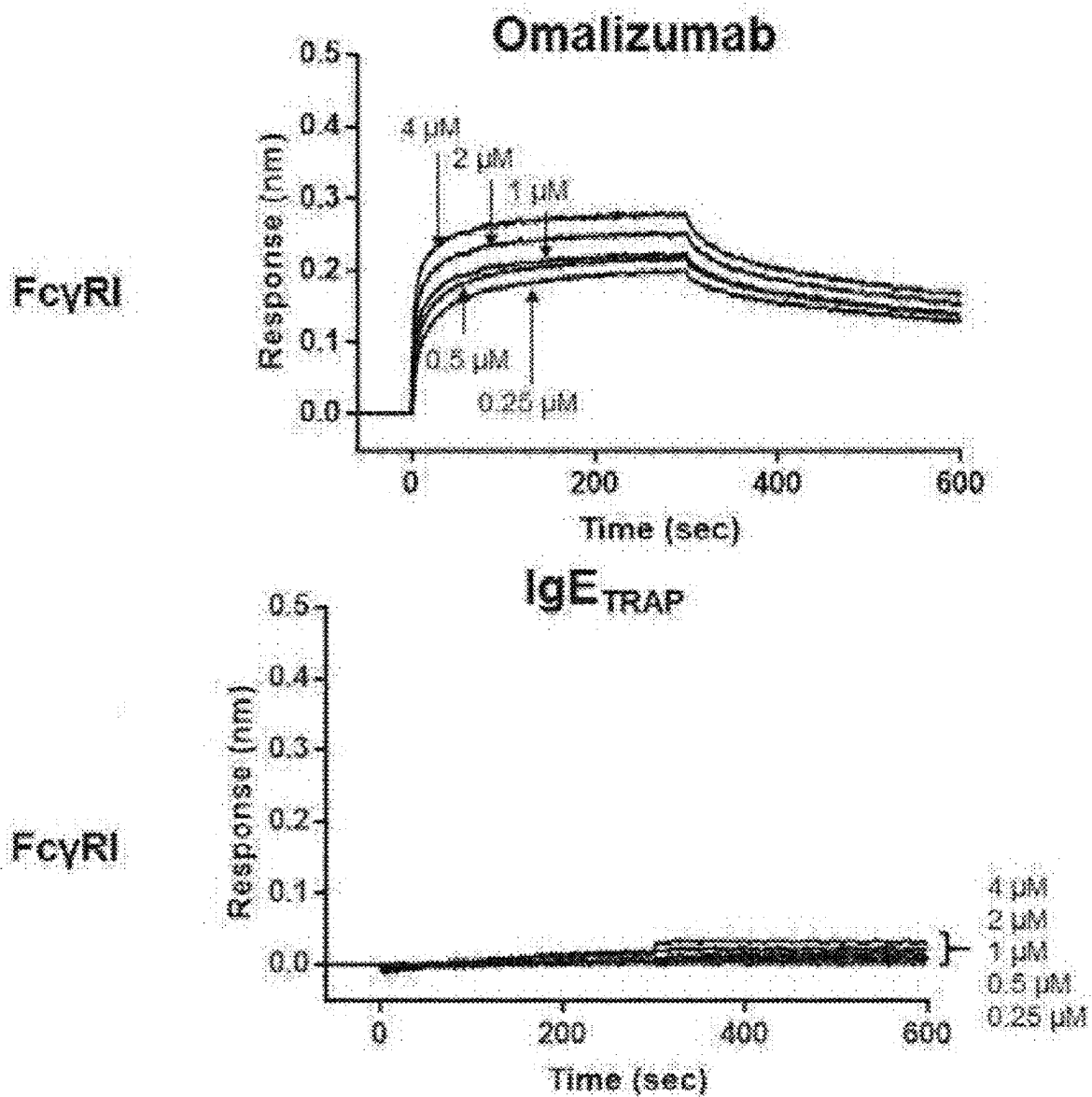


FIG.5b

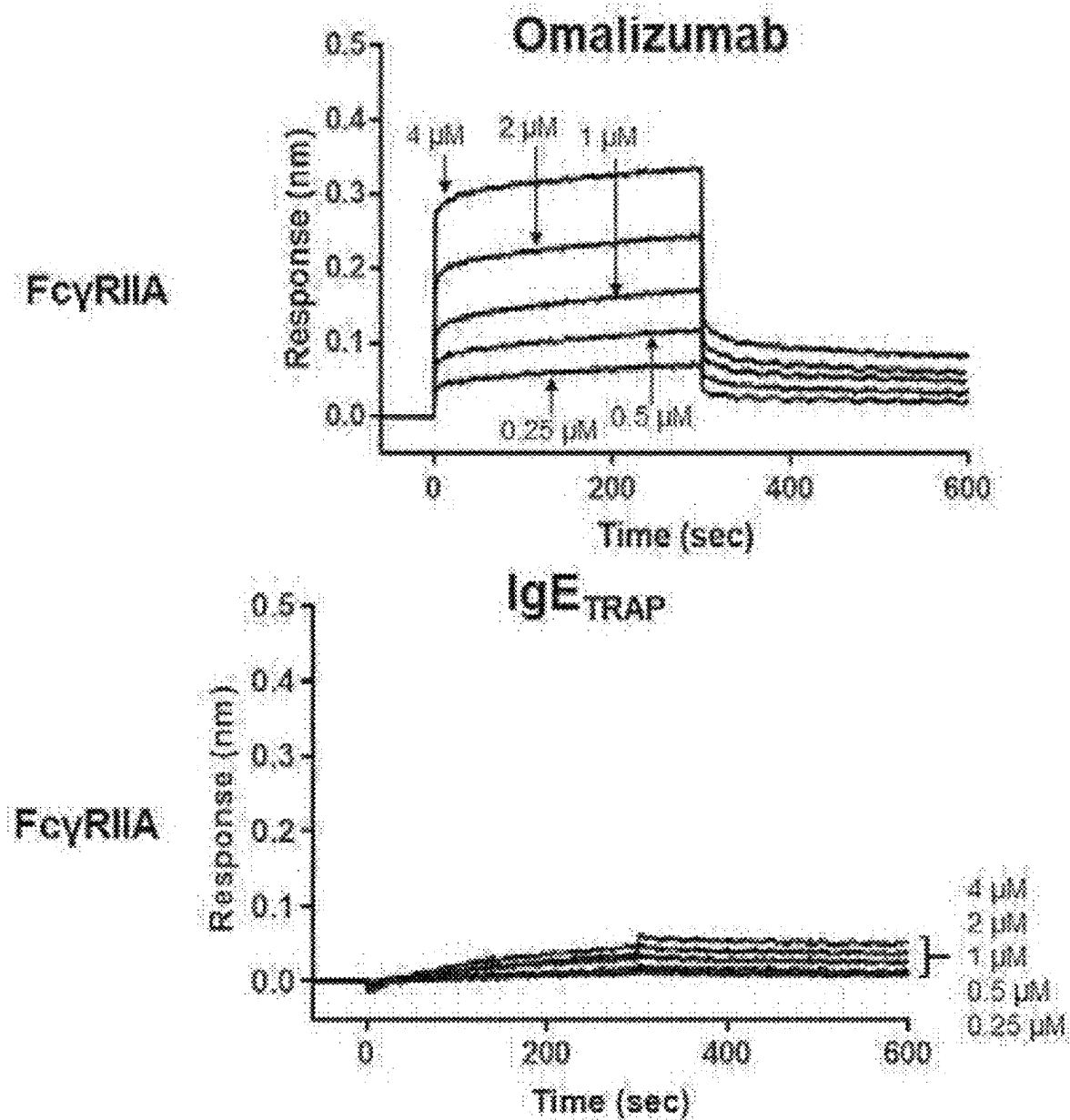


FIG.5c

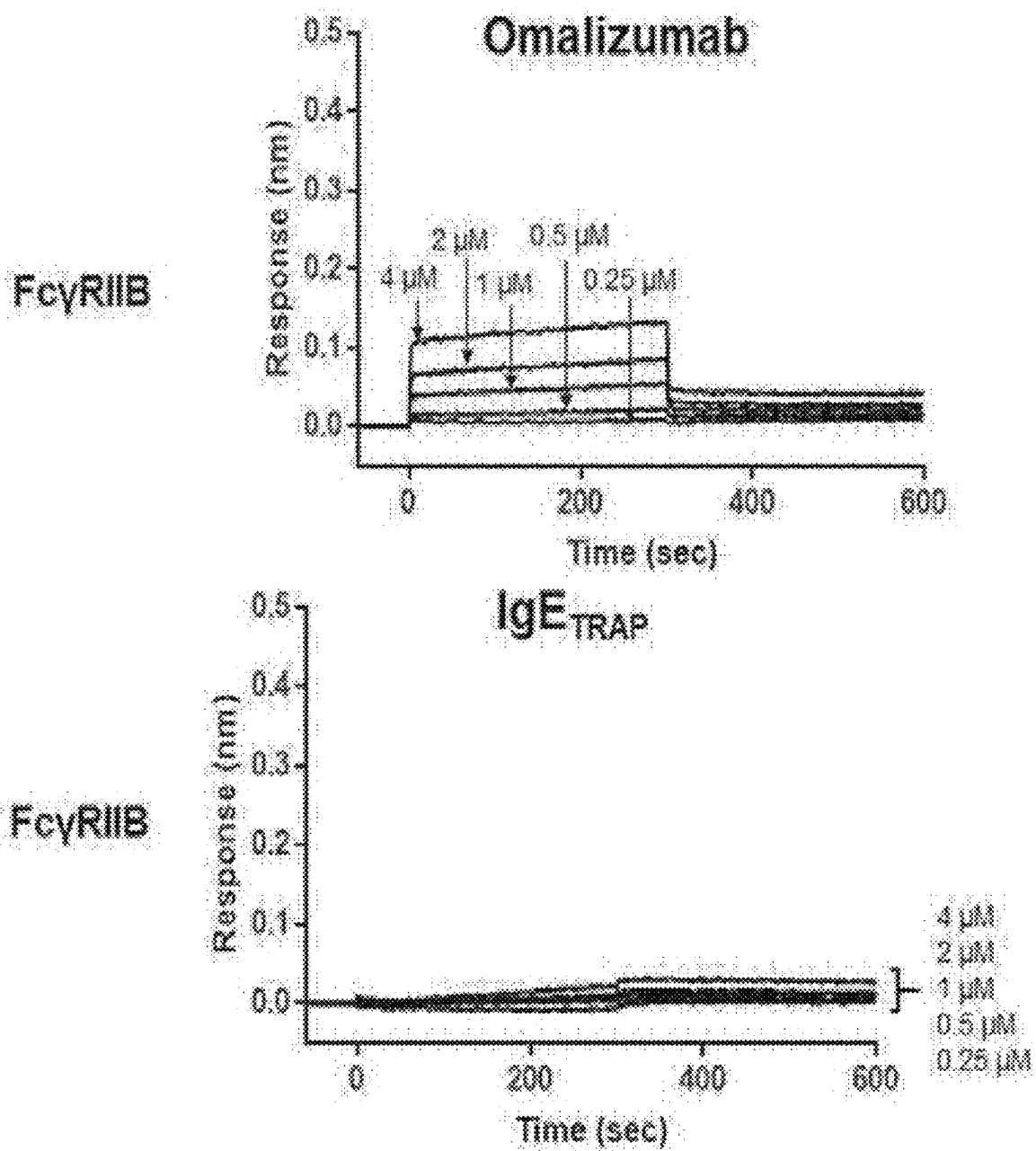


FIG. 5d

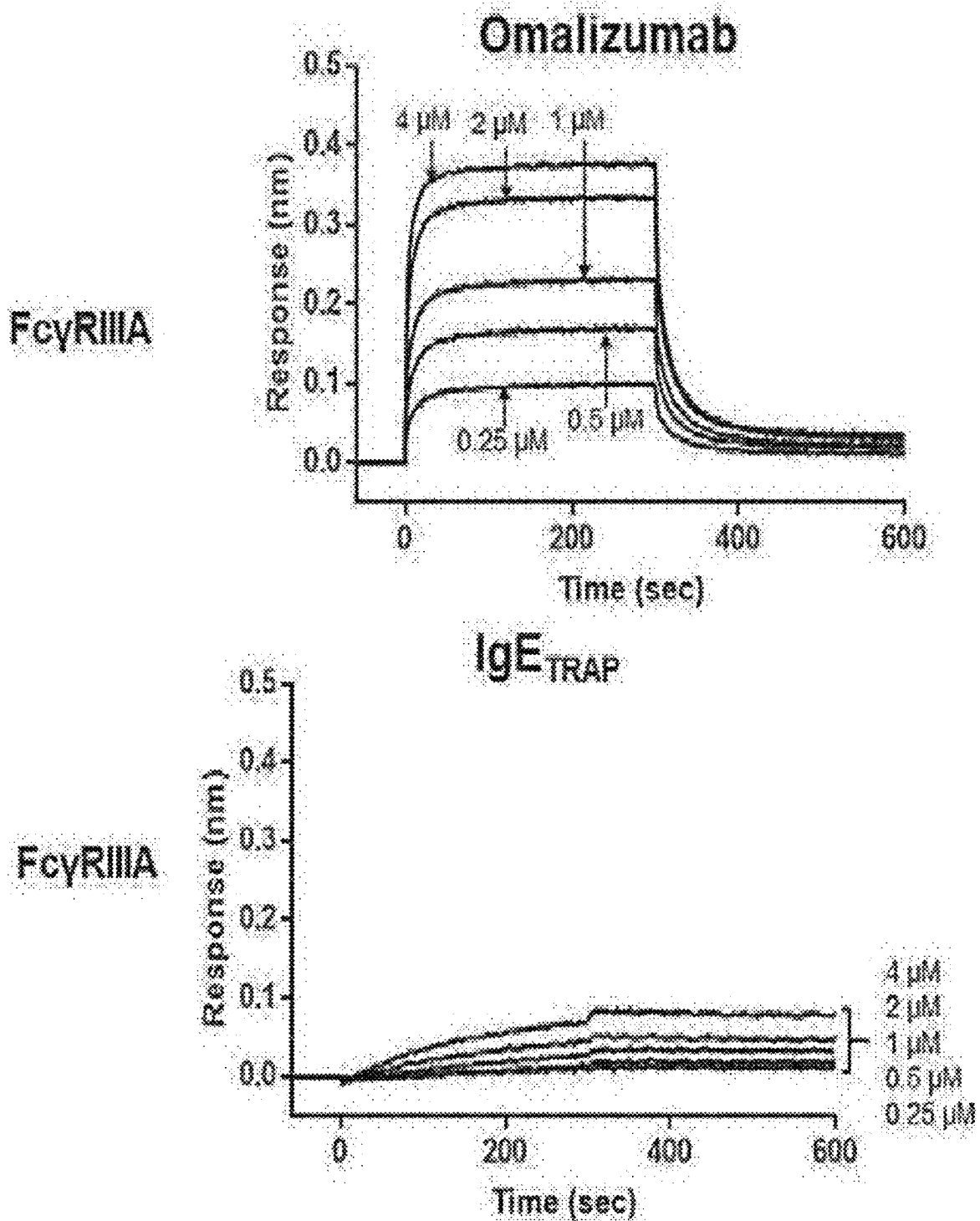


FIG.5e

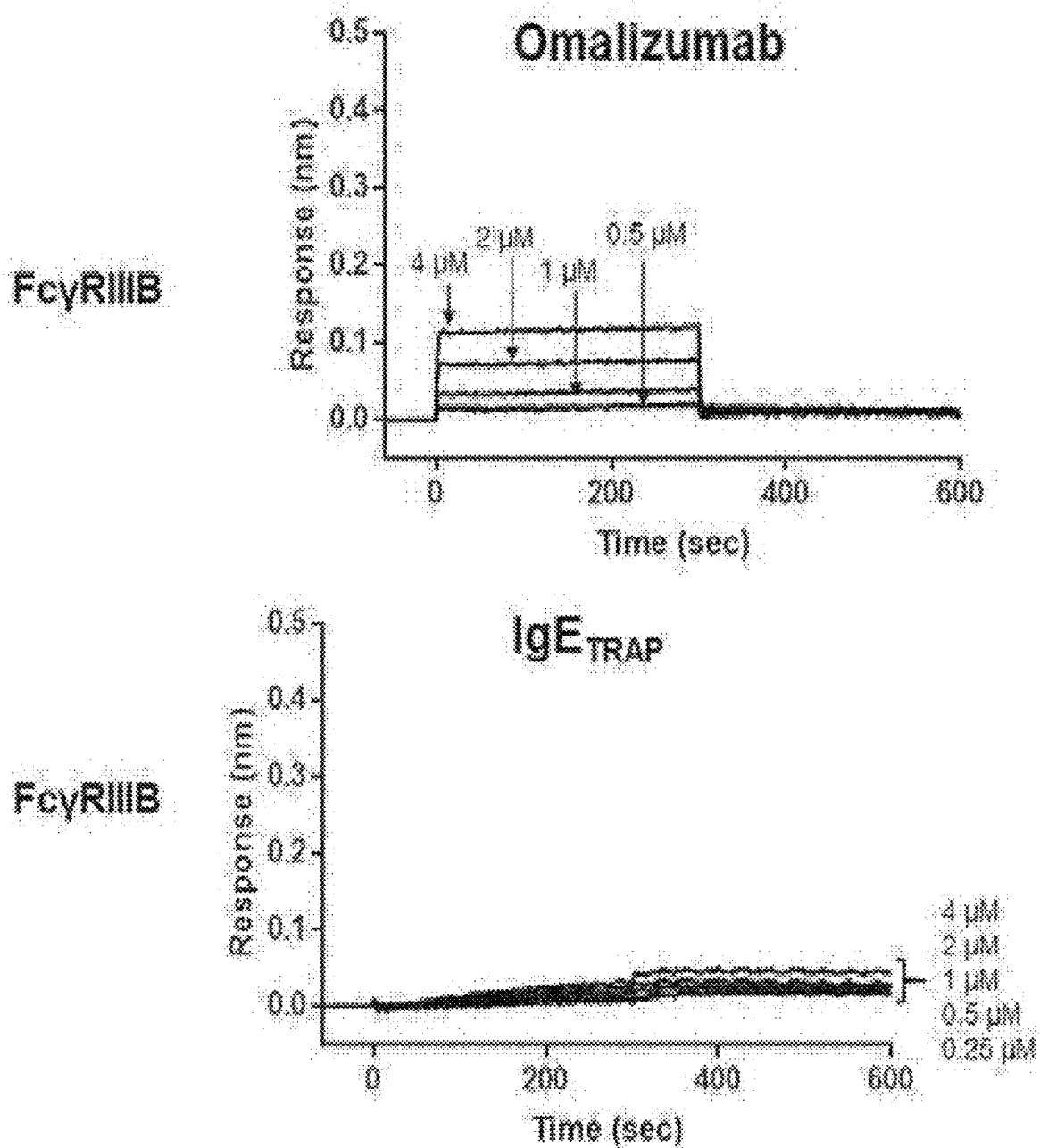


FIG. 6

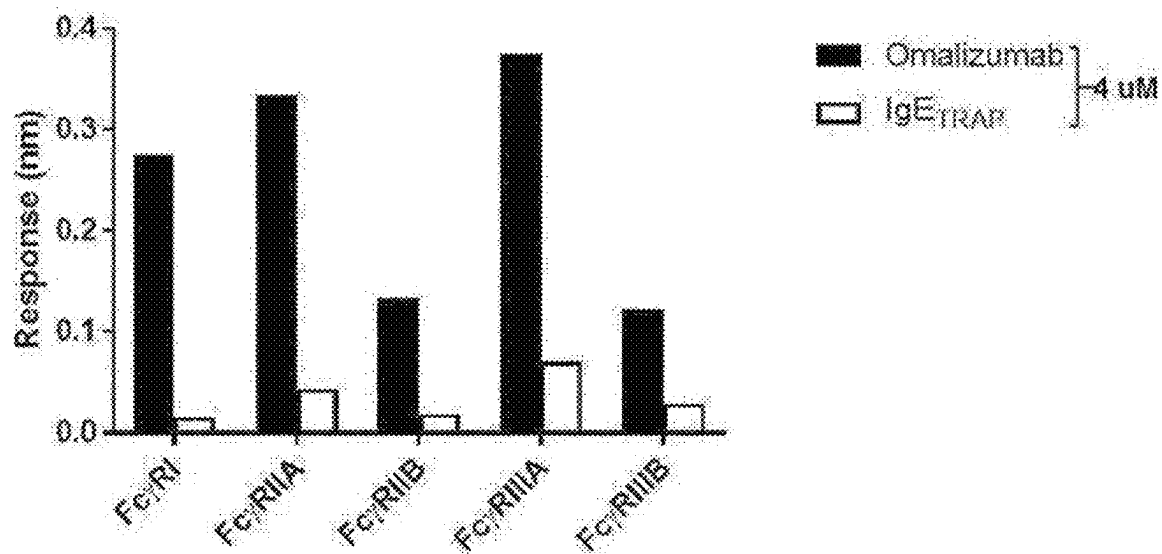


FIG. 7

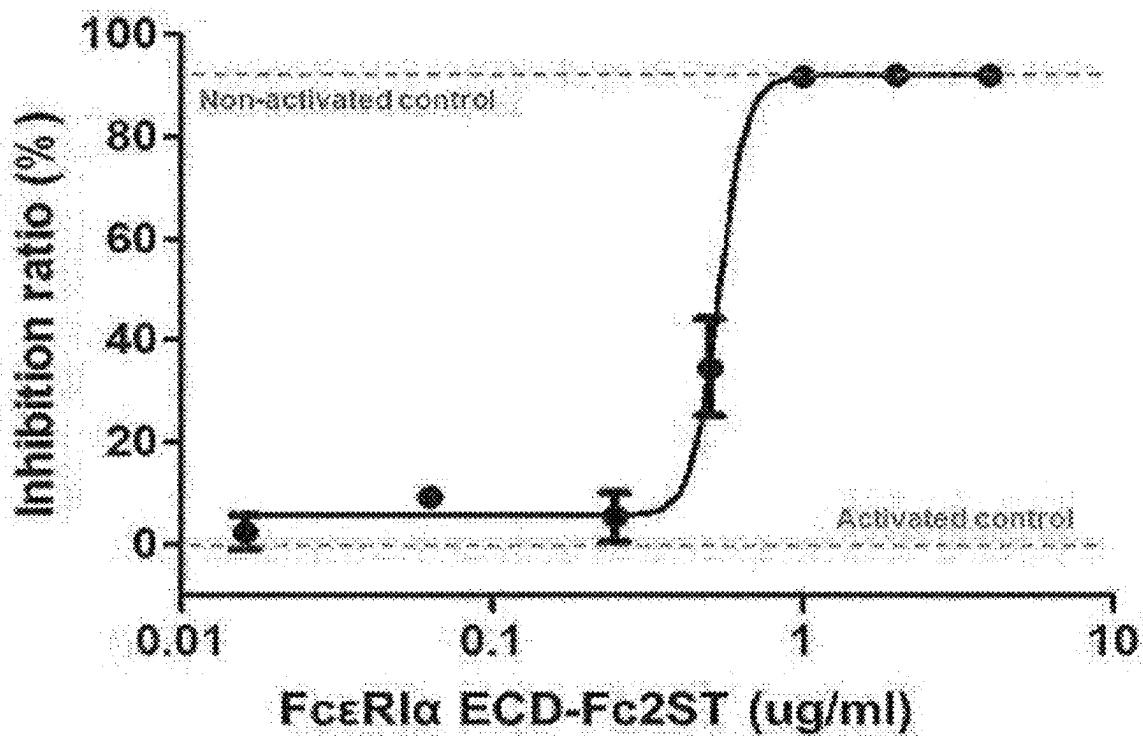


FIG.8

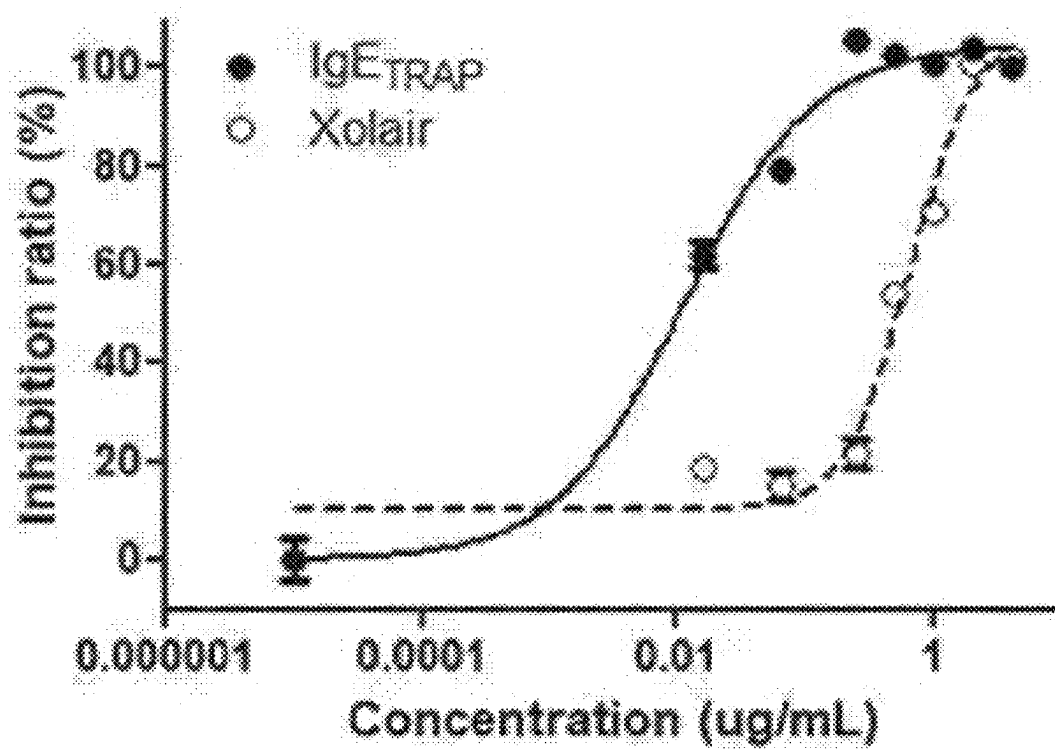


FIG.9

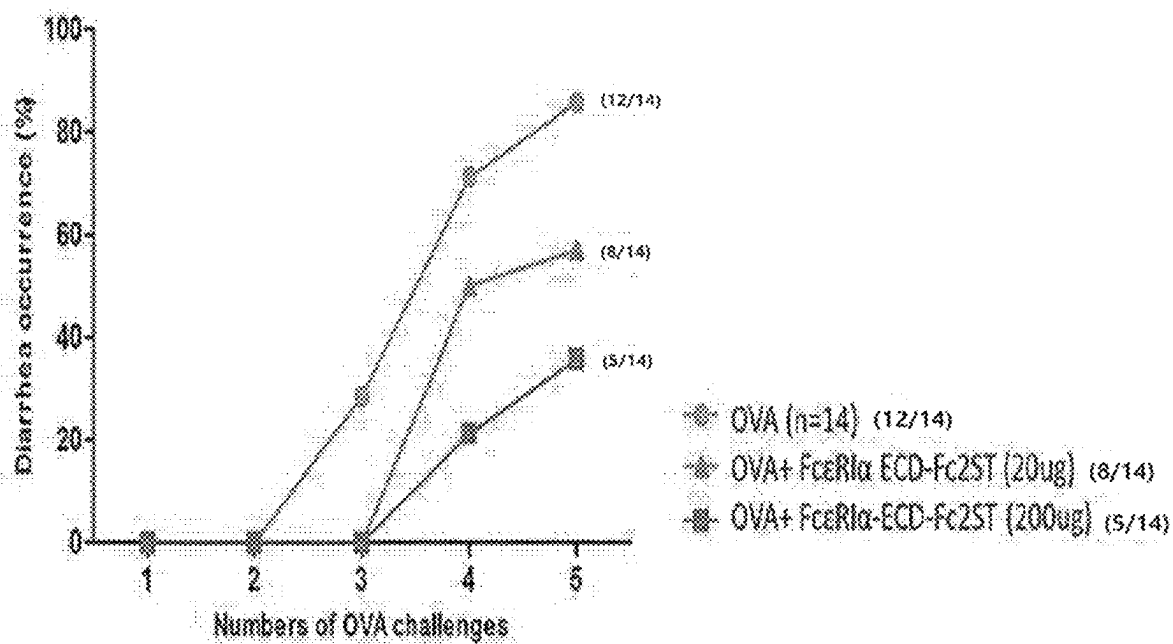


FIG.10

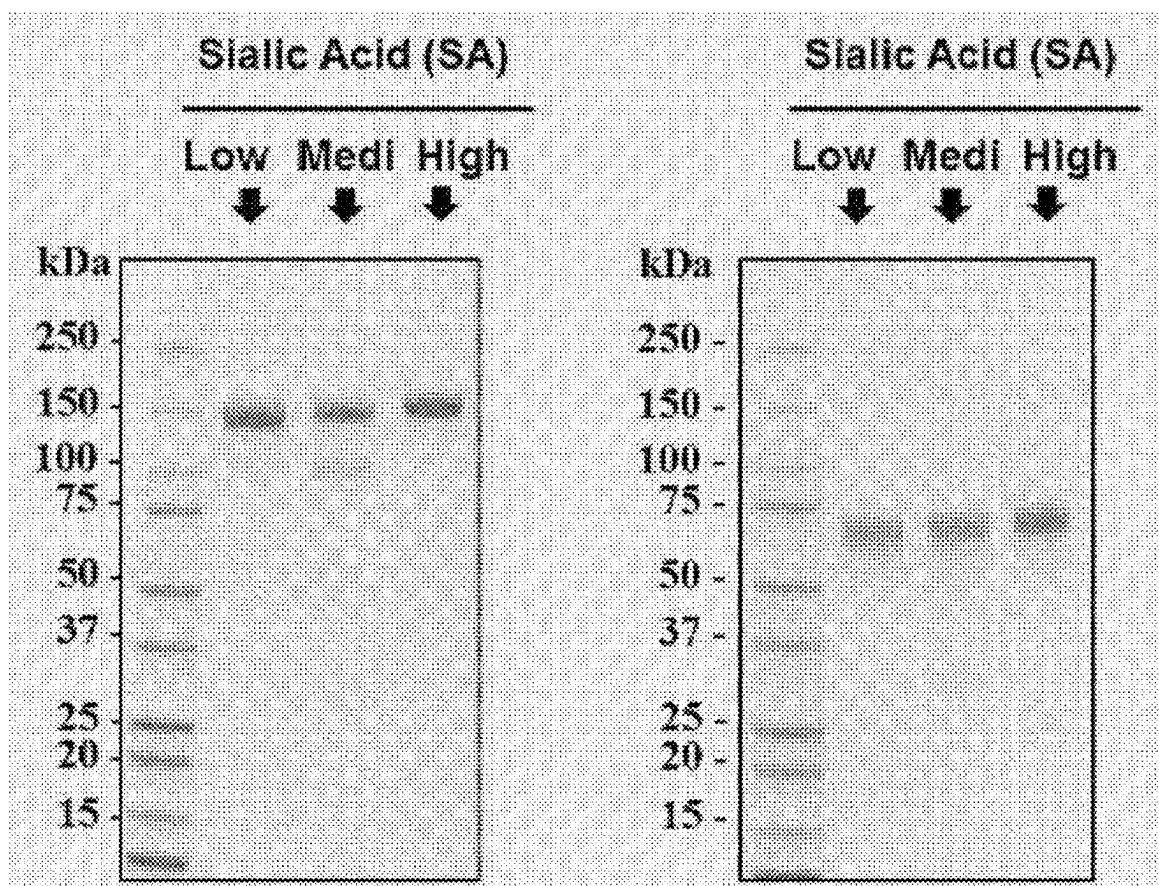


FIG.11

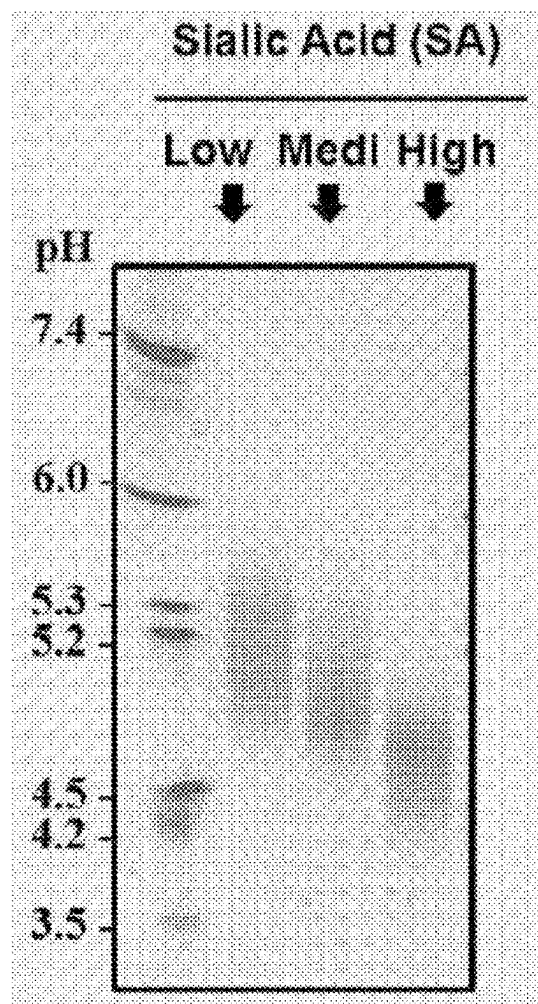


FIG.12

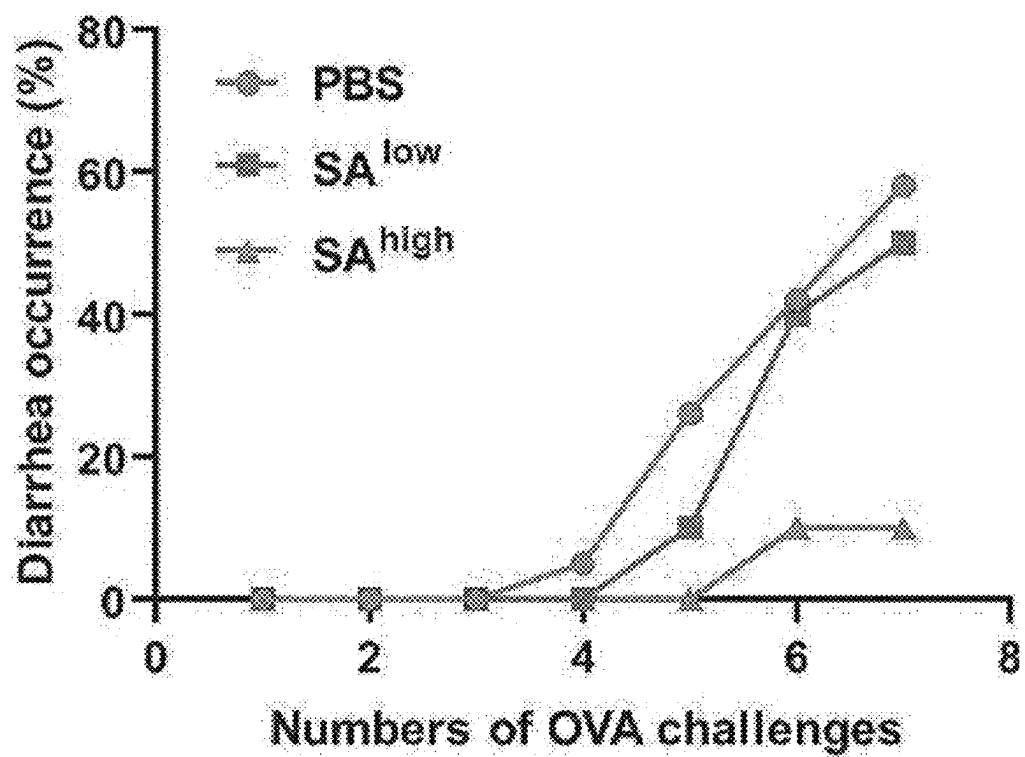


FIG.13

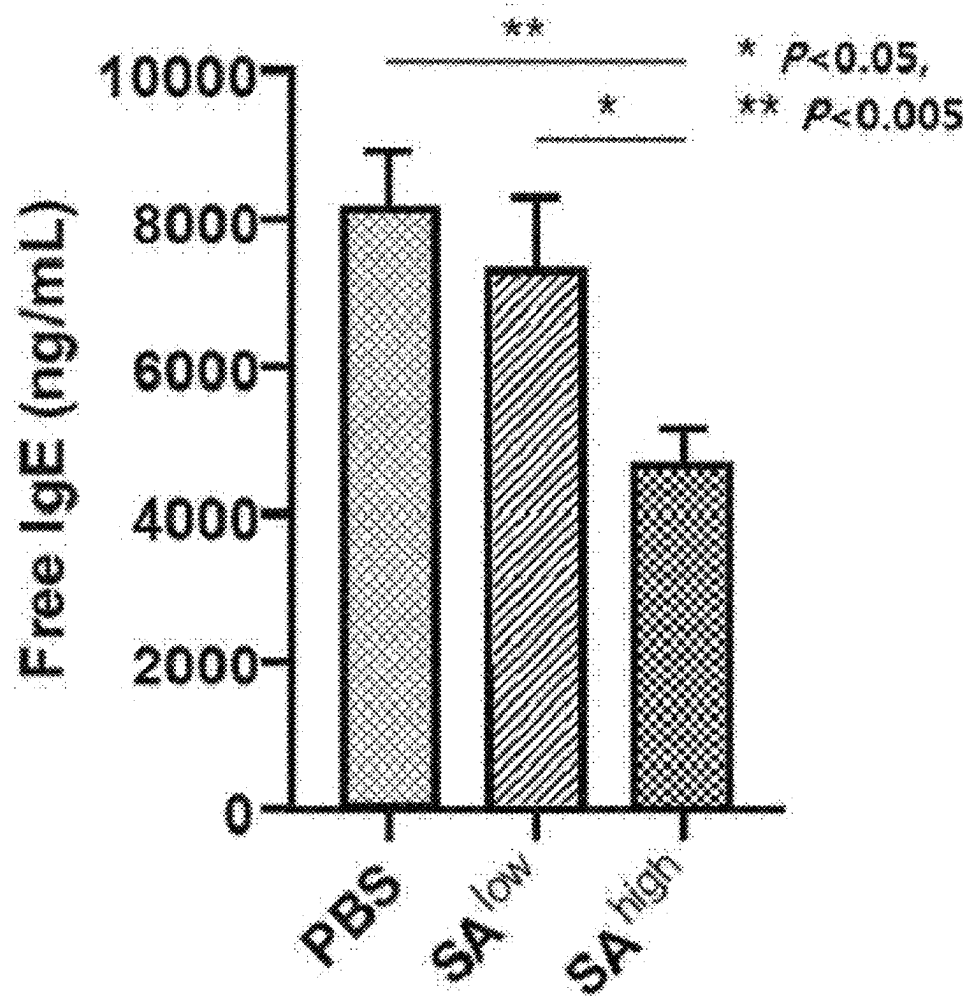


FIG.14

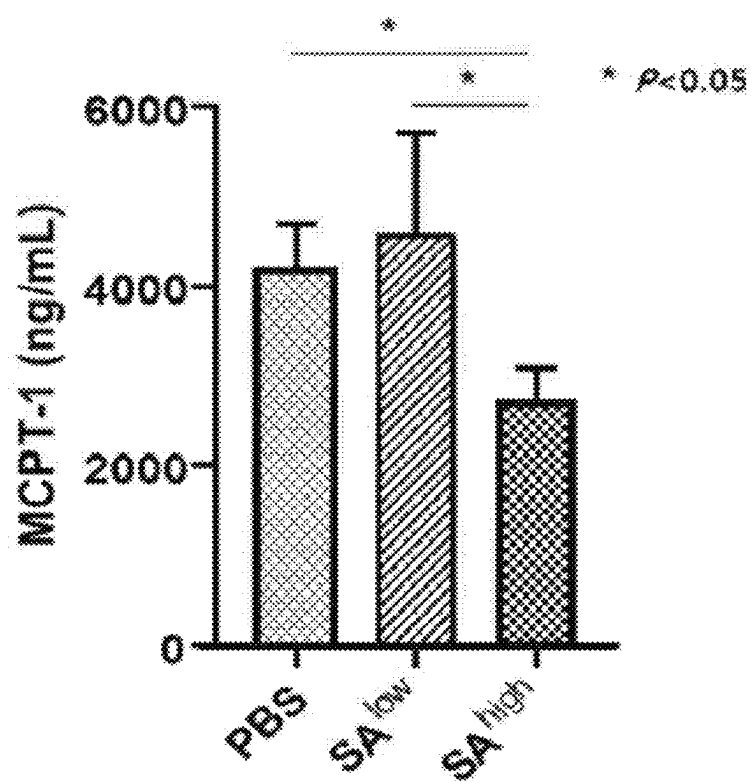


FIG.15

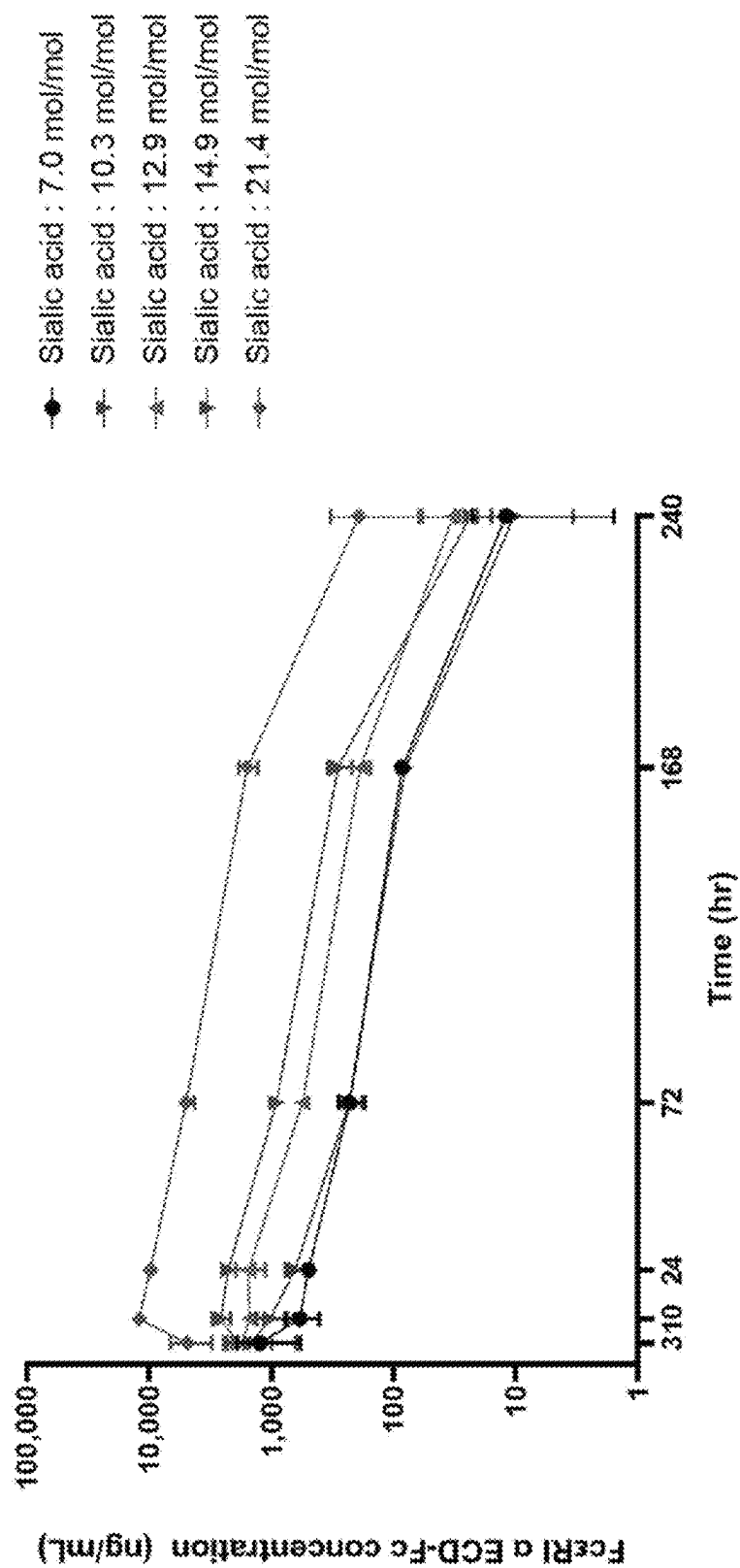
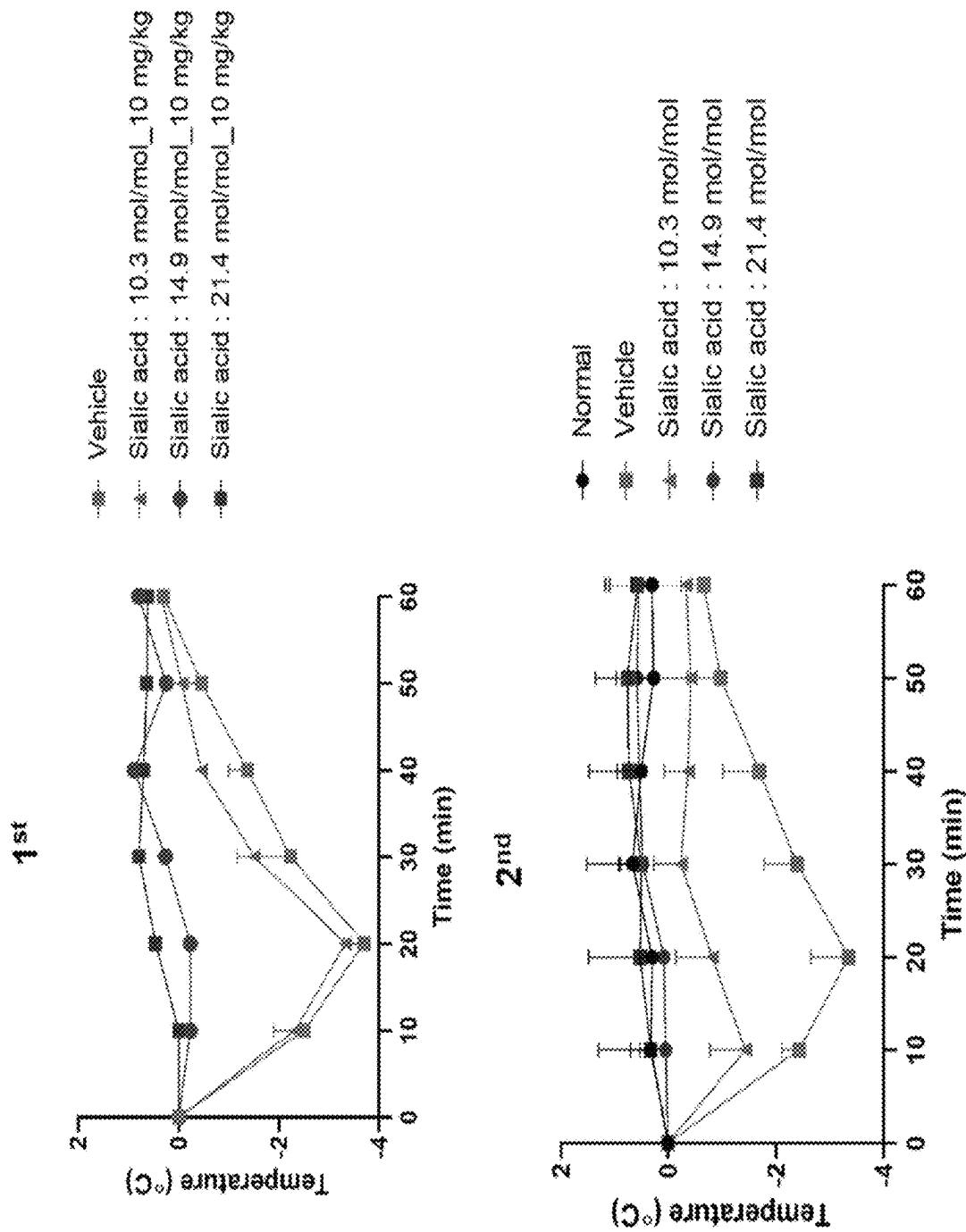


FIG.16



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**POLYPEPTIDE DIMER WITH HIGH SIALIC
ACID CONTENT, COMPRISING
EXTRACELLULAR DOMAIN OF ALPHA
SUBUNIT OF IGE FC RECEPTOR, AND
PHARMACEUTICAL COMPOSITION
COMPRISING SAME**

**CROSS REFERENCE TO RELATED
APPLICATIONS**

This application is a National Stage of International Application No. PCT/KR2020/008855 filed Jul. 7, 2020, claiming priority based on Korean Patent Application No. 10-2019-0082217 filed Jul. 8, 2019.

SEQUENCE LISTING

The content of the electronically submitted sequence listing, file name: Sequence_Listing_As_Filed.txt; file size: 36,504 bytes, date of creation: Jan. 1, 2022, filed herewith, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present invention relates to a polypeptide dimer comprising an IgE Fc receptor and having a high content of sialic acid and a pharmaceutical composition comprising the same.

BACKGROUND ART

As the modern society is industrialized and eating habits are westernized, the incidence rate of allergic diseases, such as allergic rhinitis, atopic dermatitis, and food allergy, including asthma is increasing. Development of anaphylaxis, a severe allergic disease, is also increasing. These chronic immune diseases severely impair individuals' quality of life and socioeconomic costs are soaring accordingly. Thus, there is a desperate need for measures to overcome such diseases.

Most allergic diseases are caused by an excessive immune response of immunoglobulin E (IgE). IgE is an antibody that is present in blood at a very low concentration under a normal condition. IgE is also usually produced by innocuous antigens. There is a case where the number of IgE is increased without any particular stimulus. Such a case may lead to allergic diseases. The abnormally increased number of IgE can bind to high-affinity IgE Fc receptors (FcεR1s) which are expressed on the surface of mast cells, basophils, and the like. Such binding causes mast cells or basophils to release chemical mediators such as histamine, leukotriene, prostaglandin, bradykinin, and platelet-activating factors. Release of these chemical mediators results in allergic symptoms. In particular, allergic diseases may exhibit worsened symptoms due to the binding between IgE and FcεRI. In addition, FcεRI-expressing cells are known to increase in allergic patients.

Currently, various methods, such as allergen avoidance, administration of anti-allergic drugs, modulation of IgE synthesis in the body, and development of anti-IgE antibodies, have been proposed to treat allergic diseases. However, there still remain many drawbacks, such as insufficient drug efficacy and occurrence of serious side effects. Recently, a composition capable of binding to IgE and FcγRIIb with high affinity and inhibiting cells that express membrane-anchored IgE has been studied. This composition has been

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reported to be useful for treating IgE-mediated diseases including allergy and asthma (KR10-1783272B).

In particular, omalizumab (trade name: Xolair), which targets an Fc portion of an IgE antibody, has been developed and used as a therapeutic agent for intractable severe asthma and intractable urticaria. However, a high-dose administration of omalizumab to maintain therapeutic effects leads to a high cost burden, and side effects such as angioedema and anaphylactic reaction (The Journal of Clinical Investigation Volume 99, Number 5, March 1997, 915-925).

Besides, from the post-marketing results, serious adverse reactions such as allergic granulomatous vasculitis and idiopathic severe thrombocytopenia have been reported.

**DETAILED DESCRIPTION OF THE
INVENTION**

Technical Problem

Accordingly, the present inventors studied to develop a safe and effective therapeutic agent for allergic diseases. As a result, it was found that when the sialic acid content of a polypeptide dimer comprising two monomers (FcεRIα-ECD) comprising an extracellular domain of an alpha subunit of an IgE Fc receptor is high, the above polypeptide dimer has excellent binding affinity to IgE, as well as maintains a high concentration in the blood. In particular, the present invention was completed by confirming that even when the polypeptide dimer having a high content of sialic acid is administered subcutaneously, effective delivery into the body is possible.

Solution to Problem

An aspect of the present invention provides a polypeptide dimer comprising two monomers comprising an extracellular domain (FcεRIα-ECD) of an alpha subunit of an IgE Fc receptor, wherein the molar ratio of sialic acid/polypeptide dimer is at least 8.

Another aspect of the present invention provides a pharmaceutical composition for preventing or treating allergic diseases, comprising the polypeptide dimer having a high content of sialic acid.

Another aspect of the present invention provides a transdermal patch comprising the pharmaceutical composition.

Another aspect of the present invention provides a topical patch comprising the pharmaceutical composition.

Another aspect of the present invention provides a food composition for ameliorating or alleviating allergic symptoms, comprising the polypeptide dimer having a high content of sialic acid.

Another aspect of the present invention provides a use of the polypeptide dimer having a high content of sialic acid for preventing or treating allergic diseases.

Effect of the Invention

The polypeptide dimer having a high content of sialic acid according to the present invention not only has excellent safety and persistence in the body as compared with conventionally used anti-IgE antibodies, but also binds to IgE very strongly, which has the advantage of an extended administration cycle. In addition, the polypeptide dimer having a high content of sialic acid according to the present invention is an IgE single target substance, and unlike conventional anti-IgE antibodies to which the Fc of IgG1 is applied, does not bind to an Fc gamma receptor. Thus, it can

inhibit release of mediators caused by being bound to the Fc gamma receptor on the surface of mast cells, so that severe side effects such as occurrence of anaphylaxis which can be caused by binding between IgG1 and Fc gamma receptor III on mast cells can be minimized. In addition, the polypeptide dimer having a high content of sialic acid can maintain a high concentration in the blood even in the case of subcutaneous administration. Accordingly, the polypeptide dimer having a high content of sialic acid according to the present invention can be usefully used for the prevention or treatment of allergic diseases.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a view showing the results of confirming a polypeptide dimer produced in each cell line by SDS-PAGE.

FIG. 2 is a view showing SDS-PAGE results for a non-reduced form and a reduced form of a polypeptide dimer according to one embodiment.

FIG. 3 is a view showing a binding affinity of omalizumab to IgE.

FIG. 4 is a view showing a binding affinity of the polypeptide dimer (IgE_{TRAP}) according to an embodiment of the present invention to IgE.

FIG. 5a is a view showing a binding affinity of the polypeptide dimer (IgE_{TRAP}) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor I (FcγRI).

FIG. 5b is a view showing a binding affinity of the polypeptide dimer (IgE_{TRAP}) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor IIA (FcγRIIA).

FIG. 5c is a view showing a binding affinity of the polypeptide dimer (IgE_{TRAP}) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor IIB (FcγRIIB).

FIG. 5d is a view showing a binding affinity of the polypeptide dimer (IgE_{TRAP}) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor IIIA (FcγRIIA).

FIG. 5e is a view showing a binding affinity of the polypeptide dimer (IgE_{TRAP}) according to an embodiment of the present invention or omalizumab to IgG Fc gamma receptor IIIB (FcγRIIB).

FIG. 6 is a graph obtained by quantifying a binding affinity between the polypeptide dimer (IgE_{TRAP}) according to an embodiment of the present invention or omalizumab and IgG Fc gamma receptors.

FIG. 7 is a view showing a degree of binding with IgE depending on the concentration of a polypeptide dimer according to an embodiment of the present invention.

FIG. 8 is a view showing a comparison between inhibitory abilities, on activity of human FcεRI-expressing mouse-derived mast cells, of the polypeptide dimer (IgE_{TRAP}) according to an embodiment of the present invention and Xolair (omalizumab) depending on concentrations thereof.

FIG. 9 is a view confirming the anti-allergic effect of a polypeptide dimer according to an embodiment of the present invention by measuring the frequency of diarrhea in a food allergy model.

FIG. 10 is a view showing SDS-PAGE results for a non-reduced form and a reduced form of a polypeptide dimer according to an embodiment of the present invention (SA^{low}, SA^{medi} and SA^{high}).

FIG. 11 is a view confirming the isoelectric point of a polypeptide dimer according to an embodiment of the present invention (SA^{low}, SA^{medi} and SA^{high}).

FIG. 12 is a view confirming the anti-allergic effect of subcutaneous injection of a polypeptide dimer according to an embodiment of the present invention (SA^{low} and SA^{high}) by measuring the frequency of diarrhea in a food allergy model.

FIG. 13 is a view confirming the anti-allergic effect of a polypeptide dimer according to an embodiment of the present invention (SA^{low} and SA^{high}) by measuring the IgE concentration in the blood in a food allergy model.

FIG. 14 is a view confirming the anti-allergic effect of a polypeptide dimer according to one embodiment (SA^{low} and SA^{high}) by measuring the MCPT-1 concentration in the blood in a food allergy model.

FIG. 15 is a view showing the results of analysis of a pharmacokinetic profile of a polypeptide dimer according to one embodiment depending on sialic acid content in a mouse model.

FIG. 16 is a view confirming the anti-allergic effect of a polypeptide dimer according to one embodiment depending on sialic acid content by measuring the body temperature in a passive systemic anaphylaxis mouse model.

DETAILED DESCRIPTION OF INVENTION

An aspect of the present invention provides a polypeptide dimer having a high content of sialic acid, which comprises two monomers comprising an extracellular domain (FcεRIα-ECD) of an alpha subunit of an IgE Fc receptor, wherein the monomer comprises an Fc region, and the Fc region and FcεRIα-ECD are linked via a hinge, and the polypeptide dimer is characterized in that the molar ratio of sialic acid/polypeptide dimer is at least 8.

As used herein, the term "IgE" means an antibody known as immunoglobulin E. IgE has an affinity to mast cells, basophils, or the like. In addition, reaction between an IgE and an antigen (allergen) corresponding thereto causes an inflammatory reaction. In addition, IgE is known to be a major cause of anaphylaxis.

As used herein, the term "IgE Fc receptor" is also referred to as Fcε receptor and indicates a receptor binding to an Fc portion of IgE. There are two types for the receptor. The receptor having high affinity to IgE Fc is called Fcε receptor I (FcεRI). The receptor having low affinity to IgE Fc is called Fcε receptor II (FcεRII). FcεRI is expressed in mast cells and basophils. In a case where IgE antibodies bound to FcεRI are cross-linked by polyvalent antigens, degranulation occurs in mast cells or basophils, thereby releasing various chemical transmitter substances including histamine. This release leads to an immediate allergic reaction.

The FcεRI is a membrane protein composed of one α chain, one β chain, and two γ chains linked by a disulfide bond. Among these chains, a portion to which IgE binds is the α chain (FcεRIα). FcεRIα has a size of about 60 kDa, and is composed of a hydrophobic domain existing inside the cell membrane and a hydrophilic domain existing outside the cell membrane. In particular, IgE binds to an extracellular domain of the α chain. The FcεRIα can be interchangeably used as the alpha subunit of the IgE Fc receptor.

Specifically, the alpha subunit of the IgE Fc receptor may have the amino acid sequence set forth in NP_001992.1. In addition, the extracellular domain (FcεRIα-ECD) of the alpha subunit of the IgE Fc receptor may have the amino acid sequence of SEQ ID NO: 1. The FcεRIα-ECD may be a fragment or variant of the FcεRIα-ECD, as long as the fragment or variant is capable of binding to IgE. In addition,

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the FcεRIα-ECD of SEQ ID NO: 1 may be encoded by a polynucleotide having the sequence of SEQ ID NO: 5.

The variant may be prepared through a method of substituting, deleting, or adding one or more amino acids in the wild-type FcεRIα-ECD, as long as the method does not alter a function of the FcεRIα-ECD. Such variant may be 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or more identical to the amino acid sequence of SEQ ID NO: 1.

Here, the Fc region may be the wild-type Fc region or a modified Fc region. In addition, as used herein, the term "modified Fc region" means a region in which a part of an Fc portion of an antibody has been modified. Here, the Fc region refers to a protein which contains heavy chain constant region 2 (CH2) and heavy chain constant region 3 (CH3) of an immunoglobulin, and does not contain variable regions of heavy and light chains and light chain constant region 1 (CH1) of an immunoglobulin. In particular, the modified Fc region may be obtained by substituting some amino acids in the Fc region or by combining different types of Fc regions. Specifically, the modified Fc region may have the amino acid sequence of SEQ ID NO: 2. In addition, the modified Fc region of SEQ ID NO: 2 may be encoded by a polynucleotide having the sequence of SEQ ID NO: 6.

In addition, the modified Fc region may be sugar chains in a native form, or increased sugar chains relative to a native form. Immunoglobulin Fc sugar chains may be modified by conventional methods such as chemical methods, enzymatic methods, and genetic engineering methods using microorganisms. In addition, the modified Fc region may be a region that lacks antibody dependent cellular cytotoxicity (ADCC) and complement dependent cytotoxicity (CDC) functions due to having no binding site for Fcγ receptors (FcγR) or complement component 1q (C1q).

In an embodiment, the modified Fc region and the FcεRIα-ECD may be linked via a hinge of an IgD. The hinge may be a hinge region derived from immunoglobulin IgD or a variant thereof. The hinge region of the native IgD is composed of 64 amino acids. The hinge region derived from the immunoglobulin IgD or a variant thereof may be composed of 20 to 60 consecutive amino acids, 25 to 50 consecutive amino acids, or 30 to 40 amino acids. Here, the hinge variant may be obtained by modifying some in the amino acid sequence of the hinge region of IgD in order to minimize generation of truncated forms during a protein production process.

In an embodiment, the hinge region derived from the immunoglobulin IgD or a variant thereof may be composed of 30 or 49 amino acids. In addition, the hinge region derived from the immunoglobulin IgD or a variant thereof may contain at least one cysteine.

In an embodiment, the hinge region derived from the immunoglobulin IgD or a variant thereof may contain the following sequence:

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa1 Xaa2 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro (SEQ ID NO: 17), where Xaa1 may be Lys or Gly, and Xaa2 may be Glu, Gly, or Ser. Specifically, the hinge region derived from the immunoglobulin IgD or a variant thereof may have the amino acid sequence of SEQ ID NO: 3 or SEQ ID NO: 19, thereby minimizing generation of truncated forms during a protein production process.

In another embodiment, the hinge region derived from the immunoglobulin IgD or a variant thereof may contain the following sequence:

Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr Ala Pro Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa3 Xaa4 Lys Glu Lys Glu Glu Gln Glu Glu Arg

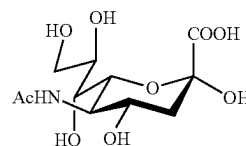
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Glu Thr Lys Thr Pro Glu Cys Pro (SEQ ID NO: 18), where Xaa3 may be Lys or Gly, and Xaa4 may be Glu, Gly, or Ser. Specifically, the hinge region derived from the immunoglobulin IgD or a variant thereof may have the amino acid sequence of SEQ ID NO: 4, thereby minimizing generation of truncated forms during a protein production process.

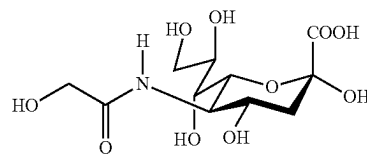
In particular, in the hinge region derived from the immunoglobulin IgD or a variant thereof having the sequence of SEQ ID NO: 4, at least one of Thr's may be glycosylated. Specifically, among the amino acids of SEQ ID NO: 18, the 13th, 14th, 18th, or 19th Thr's may be glycosylated. Preferably, all of the four Thr's may be glycosylated. Here, the glycosylation may be O-glycosylation.

The sugar chain attached to the glycoprotein pharmaceuticals is one of the main factors that determine the quality to play an important role in therapeutic efficacy, persistence in the body, targeting and immune response and the like. It was confirmed that a polypeptide dimer in which the end of sugar chain is not terminated with sialic acid was rapidly removed from the body.

The term "sialic acid" used in the present invention may include N-acetylneuraminic acid (Neu5Ac) of Formula 1 below and N-glycolylneuraminic acid (Neu5Gc) of Formula 2 below.



[Formula 1]



[Formula 2]

In this case, in one embodiment, a polypeptide dimer having a high content of sialic acid may have a high content of N-acetylneuraminic acid.

In addition, a content of sialic acid of a polypeptide dimer may be increased through a purification method. In addition, the content of sialic acid of the polypeptide dimer may be increased by producing a polypeptide dimer in a cell into which a sialic acid transferase gene is introduced.

The polypeptide dimer having a high content of sialic acid may be characterized in that the molar ratio of sialic acid/polypeptide dimer is 8 or more. For example, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29 or 30. Preferably, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25.

In an embodiment, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be at least 8, at least 9, at least 10, at least 11, at least 12, at least 13, at least 14, at least 15, at least 16, at least 17, at least 18, at least 19, at least 20, or at least 21.

In addition, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 8 to 30 or 12 to 25. In addition, the molar ratio of

sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 10 to 25, 11 to 24 or 12 to 23. Also, the molar ratio of sialic acid/polypeptide dimer in the polypeptide dimer having a high content of sialic acid may be 13 to 22, 14 to 22, 15 to 22, 16 to 22, 17 to 22, 18 to 22, or 19 to 22. Here, the sialic acid may be N-acetylneuraminic acid.

In addition, the sialic acid can bind to 8 positions on a monomer comprising an extracellular domain (FcεRIα-ECD) of an alpha subunit of an IgE Fc receptor and an Fc region. In this case, the sialic acid can bind to at least 4 or more of the above positions. Specifically, the sialic acid can bind to 4, 5, 6, 7 or 8 positions on the monomer.

As described above, the polypeptide dimer having a high content of sialic acid provided by the present invention may be in a form in which two monomers are bound to each other and each monomer is obtained by binding between an extracellular domain of an alpha subunit of an IgE Fc receptor and a modified Fc region. The polypeptide dimer having a high content of sialic acid may be in a form in which the same two monomers are bound to each other by cysteine located at a hinge site. In addition, the polypeptide dimer having a high content of sialic acid may be in a form in which two different monomers are bound to each other.

In this case, each polypeptide monomer may comprise sialic acid at the same position, but may comprise sialic acid at a different position. In addition, if the molar ratio of sialic acid/polypeptide dimer is 8 or more, each monomer constituting the dimer may have a different molar ratio of sialic acid/polypeptide.

In addition, the polypeptide dimer may contain two monomers different from each other. Specifically, it may be in a form in which one monomer may contain the extracellular domain of the alpha subunit of the IgE Fc receptor, and the other monomer contains a fragment of the extracellular domain of the alpha subunit of the IgE Fc receptor. Here, an embodiment of the monomer may be composed of the amino acid sequence represented by SEQ ID NO: 20, SEQ ID NO: 21, or SEQ ID NO: 22.

In addition, the polypeptide dimer having a high content of sialic acid provided by the present invention exhibits a binding affinity to IgE which is 10- to 100-fold, 20- to 90-fold, 20- to 70-fold, 30- to 70-fold, or 40- to 70-fold higher than omalizumab, an anti-IgE antibody, and may preferably exhibit a binding affinity to IgE which is about 70-fold higher than omalizumab.

Another aspect of the present invention provides a pharmaceutical composition for preventing or treating allergic diseases, comprising the polypeptide dimer having a high content of sialic acid.

The polypeptide dimer is the same as described above. In particular, the pharmaceutical composition may be characterized by being for subcutaneous injection. It was confirmed that a polypeptide dimer having a high content of sialic acid is effective in treating and/or preventing allergic diseases compared to a polypeptide dimer having a low content of sialic acid upon subcutaneous injection.

As used herein, the term "allergic disease" means a pathological symptom caused by an allergic reaction mediated by mast cell activation such as mast cell degranulation. For example, such allergic diseases may be the one selected from the group consisting of food allergy, atopic dermatitis, asthma, allergic rhinitis, allergic conjunctivitis, allergic dermatitis, chronic idiopathic urticaria and allergic contact dermatitis, but are not limited thereto. In particular, the allergic diseases may be IgE-mediated.

In the pharmaceutical composition, the polypeptide dimer may be contained in any amount (effective amount) depending on use, formulation, blending purpose, and the like, as long as the polypeptide dimer can exhibit anti-allergic activity. A typical effective amount of the polypeptide dimer will be determined within a range of 0.001% by weight to 20.0% by weight based on a total weight of the composition. Here, the effective amount refers to an amount which is capable of inducing an anti-allergic effect. Such an effective amount can be determined by those ordinarily skilled in the art.

The pharmaceutical composition may further contain a pharmaceutically acceptable carrier. Specifically, the pharmaceutical composition contains a pharmaceutically acceptable carrier, and may be made into an oral or parenteral formulation depending on a route of administration by a conventional method known in the art.

In a case where the pharmaceutical composition of the present invention is made into an oral formulation, the pharmaceutical composition may be made into formulations such as powders, granules, tablets, pills, sugar coating tablets, capsules, liquids, gels, syrups, suspensions, and wafers, together with suitable carriers, in accordance with methods known in the art. Here, examples of suitable pharmaceutically acceptable carriers can include sugars such as lactose, glucose, sucrose, dextrose, sorbitol, mannitol, and xylitol, starches such as corn starch, potato starch, and wheat starch, celluloses such as cellulose, methylcellulose, ethylcellulose, sodium carboxymethyl cellulose, and hydroxypropylmethyl cellulose, polyvinyl pyrrolidone, water, methylhydroxybenzoate, propylhydroxybenzoate, magnesium stearate, mineral oil, malt, gelatin, talc, polyol, vegetable oil, and the like. In a case of being made into preparations, the preparations can be carried out, as necessary, by including diluents and/or excipients such as a filler, an extender, a binder, a wetting agent, a disintegrant, and a surfactant.

In a case where the pharmaceutical composition of the present invention is made into a parenteral formulation, the pharmaceutical composition may be made into preparations in the form of injections, transdermal drugs, nasal inhalers, and suppositories, together with suitable carriers, in accordance with methods known in the art. In a case of being prepared into injections, sterilized water, ethanol, polyol such as glycerol and propylene glycol, or a mixture thereof may be used as a suitable carrier. For the carrier, isotonic solutions such as Ringer's solution, phosphate buffered saline (PBS) containing triethanolamine, sterile water for injection, and 5% dextrose, and the like may be preferably used.

Preparation of the pharmaceutical composition is known in the art and specifically, reference can be made to Remington's Pharmaceutical Sciences (19th ed., 1995) and the like.

A preferable daily dosage of the pharmaceutical composition of the present invention is ranged from 0.01 μg/kg to 10 g/kg, and preferably from 0.01 mg/kg to 1 g/kg, depending on the patient's condition, body weight, sex, age, disease severity, or route of administration. The pharmaceutical composition may be administered once a day or several times a day when the dosage is high.

The subject to which the composition of the present invention can be applied (prescribed) is a mammal and may be, for example, a human, a dog and a cat, with a human being particularly preferred but not limited thereto. The pharmaceutical composition may further comprise any compound or natural extract, on which safety has already been

verified and which is known to have anti-allergic activity, for the purpose of raising and reinforcing the anti-allergic activity.

The present inventors studied to develop a safe and effective therapeutic agent for allergic diseases. As a result, it was confirmed that a polypeptide dimer having a high content of sialic acid has excellent binding affinity with IgE compared to a polypeptide dimer having a low content of sialic acid (Table 3). In addition, it was confirmed that when a polypeptide dimer having a high content of sialic acid is administered subcutaneously to a subject suffering from an allergic disease, the concentration of IgE and MCPT-1 in the blood is effectively reduced compared to a polypeptide dimer having a low content of sialic acid, and the effect persists even after a predetermined time has passed (FIGS. 13 and 14).

Accordingly, another aspect of the present invention provides a transdermal patch comprising the pharmaceutical composition. In addition, another aspect of the present invention provides a topical patch comprising the pharmaceutical composition.

Another aspect of the present invention provides a food composition for ameliorating or alleviating allergic symptom, comprising the polypeptide dimer having a high content of sialic acid.

The polypeptide dimer having a high content of sialic acid is the same as described above. In addition, the polypeptide dimer having a high content of sialic acid may be bound to appropriate delivery means for efficient delivery into the intestine.

In addition, the food composition may be prepared in any form and may be, for example, prepared in the form of beverages such as tea, juice, carbonated beverage, and ionic beverage, processed dairy products such as milk and yogurt, health functional food preparations such as tablets, capsules, pills, granules, liquids, powders, flakes, pastes, syrups, gels, jellies, and bars, or the like.

The food composition may fall within any product category in legal or functional classification as long as the food composition complies with the enforcement regulations at the time of being manufactured and distributed. For example, the food composition may be a health functional food according to the Health Functional Foods Act, or may fall within confectioneries, beans, teas, beverages, special-purpose foods, or the like according to each food type in the Food Code of Food Sanitation Act (standards and specifications for food, notified by Food and Drug Administration). With regard to other food additives that may be contained in the food composition of the present invention, reference can be made to the Food Code or the Food Additive Code according to the Food Sanitation Act.

Another aspect of the present invention provides a method for treating or preventing allergic diseases, comprising a step of administering to a subject the polypeptide dimer having a high content of sialic acid. The polypeptide dimer having a high content of sialic acid is the same as described above.

The subject may be a mammal, preferably, a human, a dog and a cat. Here, administration may be achieved orally or parenterally. Here, parenteral administration may be performed by methods such as subcutaneous administration, intravenous administration, mucosal administration, and muscular administration.

The allergic diseases may be the one selected from the group consisting of food allergy, atopic dermatitis, asthma, allergic rhinitis, allergic conjunctivitis, allergic dermatitis, chronic idiopathic urticaria and allergic contact dermatitis.

Another aspect of the present invention provides a use of the polypeptide dimer having a high content of sialic acid for preventing or treating allergic diseases.

Hereinafter, the present invention will be described in more detail with reference to the following examples. However, the following examples are intended to merely illustrate the present invention, and the scope of the present invention is not limited only thereto.

Example 1. Preparation of Polypeptide Containing FcεRIα-ECD and Fc Region

A C-terminal modified polypeptide of the extracellular domain (FcεRIα-ECD) of the alpha subunit of the IgE Fc receptor was prepared according to the method disclosed in U.S. Pat. No. 7,867,491.

First, in order to express a protein (FcεRIαECD-Fc1), a protein (FcεRIαECD-Fc2), and a protein (FcεRIαECD-Fc3), in which the extracellular domain of the α-chain of FcεRI having the amino acid sequence of SEQ ID NO: 1 and the modified immunoglobulin Fc of SEQ ID NO: 2 are linked via a hinge of SEQ ID NO: 19, a hinge of SEQ ID NO: 3, and a hinge of SEQ ID NO: 4, respectively, cassettes obtained by linking the gene encoding each protein were cloned into the pAD15 vectors (Genexin, Inc.) to construct FcεRIαECD-Fc protein expression vectors. Then, each of the expression vectors was transduced into CHO DG44 cells (from Dr. Chasm, Columbia University, USA).

Here, at the time of being transduced into the cell line, an expression vector obtained by cloning an α-2,6-sialic acid transferase gene into the pCI Hygro vector (Invitrogen) was simultaneously transduced to separately prepare cell lines which are capable of expressing FcεRIαECD-Fc2ST and FcεRIαECD-Fc3ST proteins to which sialic acid is added.

As a primary screening procedure, HT selection was carried out using 5-hydroxytryptamine (HT)-free 10% dFBS medium (Gibco, USA, 30067-334), MEMα medium (Gibco, 12561, USA, Cat No. 12561-049), and HT+ medium (Gibco, USA, 11067-030). Then, methotrexate (MTX) amplification was performed using HT-selected clones to amplify productivity using the dihydrofolate reductase (DHFR)-system.

After completion of the MTX amplification, subculture was carried out about 1 to 5 times for cell stabilization for the purpose of evaluation of productivity. Thereafter, unit productivity evaluation of the MTX-amplified cells was performed. The results are shown in Table 1 below.

TABLE 1

Version	Media	MTX concen- tration	Productivity		
			3-day culture		Batch culture
			ug/mL	ug/10 ⁶ cells	(mg/ml)
FcεRIαECD-Fc2	Ex-	500 nM	37.23	20.9	225
FcεRIαECD- Fc2 + α2,6-ST	cellDHFR	100 nM	45.4	25.1	338.2
FcεRIαECD-Fc3		2 uM	27.0	16.9	180.4
FcεRIαECD- Fc3 + α2,6-ST		1 uM	17.5	10.2	101.7

As shown in Table 1, the FcεRIαECD-Fc3 cell line exhibited productivity of 16.9 μg/10⁶ cells after the methotrexate amplification at 2 uM. On the other hand, the FcεRIαECD-Fc3 cell line (FcεRIαECD-Fc3ST) co-transduced with 2,6-sialic acid transferase exhibited productivity

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of $10.2 \mu\text{g}/10^6$ cells after the methotrexate amplification at 1 μM . In addition, the Fc ϵ RI α ECD-Fc2 cell line exhibited productivity of $20.9 \mu\text{g}/10^6$ cells under the methotrexate amplification condition at 0.5 μM . In addition, the Fc ϵ RI α ECD-Fc2 cell line (Fc ϵ RI α ECD-Fc2ST) co-transduced with 2,6-sialic acid transferase exhibited productivity of $25.1 \mu\text{g}/10^6$ cells after the methotrexate amplification at 0.1 μM . That is, it was identified that the Fc ϵ RI α ECD-Fc2 cell line co-transduced with 2,6-sialic acid transferase, which had been selected under the methotrexate amplification condition at 0.1 μM , exhibits the most excellent productivity.

The polypeptide (Fc ϵ RI α ECD-Fc) produced from the Fc ϵ RI α ECD-Fc2 cell line is referred to as "Fc ϵ RI α ECD-Fc2," and the polypeptide produced from the Fc ϵ RI α ECD-Fc2+a2,6-ST cell line is referred to as "Fc ϵ RI α ECD-Fc2ST". In addition, the polypeptide produced from the Fc ϵ RI α ECD-Fc3 cell line is referred to as "Fc ϵ RI α ECD-Fc3," and the polypeptide produced from the Fc ϵ RI α ECD-Fc3+a2,6-ST cell line is referred to as "Fc ϵ RI α ECD-Fc3ST".

Example 2. Confirmation of Purification and Purity of Polypeptide (Fc ϵ RI α ECD-Fc)

Among the cell lines selected in Example 1 above, i) Fc ϵ RI α ECD-Fc3 cell line, ii) Fc ϵ RI α ECD-Fc3+a2,6-ST cell line, and iii) Fc ϵ RI α ECD-Fc2+a2,6-ST cell line were cultured at a 60 ml scale by a batch culture method. The resulting cultures were purified using a Protein-A affinity column to obtain Fc ϵ RI α ECD-Fc, and then the purified Fc ϵ RI α ECD-Fc were subjected to SE-HPLC (Size-Exclusion High performance liquid chromatography) and SDS-PAGE to identify purity of the polypeptide.

Specifically, SDS-PAGE was performed under a non-reducing condition. In the non-reducing condition, each purified polypeptide was mixed with a non-reducing sample buffer, and then electrophoresis was performed for 30 minutes in a Mini-Protean TGX™ gels (Bio-Rad) in a TGS (Tris Glycine SDS) buffer under a 200V condition. After electrophoresis, a protein was stained with a Coomassie Brilliant Blue solution. The results are shown in Table 2 and FIG. 1 below:

TABLE 2

Lane #	Sample	Purification	Purity (SE-HPLC)	Sample condition
M	Protein standard	One-step	—	—
1	Fc ϵ RI α ECD-Fc3	(Protein-A affinity)	94.5%	—
2	Fc ϵ RI α ECD-Fc3ST	column)	93.7%	Non-reducing
3	Fc ϵ RI α ECD-Fc2ST		93.2%	
4	Fc ϵ RI α ECD-Fc3	Purification	94.5%	Freezing/Thawing test
5	Fc ϵ RI α ECD-Fc3ST		93.7%	
6	Fc ϵ RI α ECD-Fc2ST		93.2%	

As shown in Table 2 above, it was confirmed that the purity of each polypeptide purified by the SE-HPLC method was 93% or more. It was confirmed that impurities such as a truncated form did not appear in a non-reducing condition, and in particular, the purity was 93% or more and there were no impurities even after the thawing/freezing process.

Experimental Example 1. Confirmation of Dimer Formation of Polypeptide (Fc ϵ RI α ECD-Fc)

Among the cell lines selected in Example 1 above, i) Fc ϵ RI α ECD-Fc3 cell line, ii) Fc ϵ RI α ECD-Fc3+a2,6-ST

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cell line, and iii) Fc ϵ RI α ECD-Fc2+a2,6-ST cell line were cultured at a 60 ml scale by a batch culture method. Thereafter, SE-HPLC and SDS-PAGE were performed on the purified product obtained from the purification of polypeptide using the culture supernatant and Protein-A affinity column. In this case, they were performed on the culture supernatant and the purified product under a non-reducing condition and a reducing condition, respectively.

The non-reducing condition was performed in the same manner as in Example 2. On the other hand, in a reducing condition, each purified polypeptide was mixed with a reducing sample buffer comprising 2-mercaptoethanol, and then denatured at a temperature of 100° C. for 5 minutes. Thereafter, electrophoresis was performed for 30 minutes under a 200V condition using a TGS buffer in Mini-Protean TGX™ gels (Bio-Rad). After electrophoresis, a protein was stained with a Coomassie Brilliant Blue solution.

As a result, a polypeptide having a size of about 150 kDa was detected under a non-reducing condition, and a polypeptide having a size of about 75 kDa was detected under a reducing condition. Through this, it was confirmed that the polypeptide forms a dimer (FIG. 2). In particular, it was confirmed that the purity of a polypeptide dimer is high even in the culture supernatant corresponding to the input, and no polypeptide was detected in the sample (FT; Flow Through) obtained by passing the culture supernatant through an affinity column, but a polypeptide was detected in the eluted sample. In addition, no significant difference was observed even when the sample (Elute N) that was immediately neutralized with 1 M Tris buffer (pH 9.0) due to the low pH of the eluted sample was compared with the sample (Elute) that was not neutralized (FIG. 2).

In addition, it was identified that not only polypeptides with very high purity (98% or higher) are purified but also polypeptides are expressed with very high purity even in the culture supernatant. This indicates that process development steps can be simplified in developing the polypeptides, which had been expressed in the cell lines in question, into a medical product, and as a result, it is highly likely for the development cost of the medical product to be remarkably decreased.

Experimental Example 2. Confirmation of Binding Affinity of Polypeptide Dimer to IgE

A binding affinity to IgE was comparatively measured for the purified products obtained by purifying the polypeptides which had been produced in Example 1, i) Fc ϵ RI α ECD-Fc2, ii) Fc ϵ RI α ECD-Fc2ST, iii) Fc ϵ RI α ECD-Fc3, and iv) Fc ϵ RI α ECD-Fc3ST, through the method of Example 2 above, and the commercially available anti-IgE antibody, omalizumab (trade name: Xolair).

Specifically, the binding affinity to IgE was measured by coating IgE on the channel of the Protein GLC sensor chip (Bio-Rad Laboratories, Inc., Cat #176-5011), and causing omalizumab or each Fc ϵ RI α ECD-Fc protein at various concentrations to flow at a rate of 30 μL per minute. Here, the experiments were conducted by identifying zero base using 25 mM NaOH as a regeneration buffer, and then repeating the above steps. Thereafter, a binding curve was identified using a protein binding analyzer (ProteOn XPR36, Bio-Rad Laboratories, Inc., USA). The results are shown in Table 3.

TABLE 3

Samples items Drug type	FcεRIα ECD-Fc Fc fusion protein	Omalizumab Anti-IgE Ab	Remarks
Binding affinity (Association rate)	Fc2	2.14×10^5	4.05×10^5
	Fc2ST	2.64×10^5	
	Fc3	1.98×10^5	
	Fc3ST	2.40×10^5	
	Fc2	8.29×10^{-5}	6.02×10^{-3}
	Fc2ST	5.69×10^{-5}	
	Fc3	1.33×10^{-4}	
	Fc3ST	1.49×10^{-4}	
	Fc2	3.88×10^{-10}	1.49×10^{-8}
	Fc2ST	2.16×10^{-10}	
KD (kd/ka)	Fc3	6.72×10^{-10}	
	Fc3ST	6.21×10^{-10}	

As shown in Table 3, the association rate (ka) value of the polypeptide dimer according to an embodiment of the present invention and IgE was measured to be 1.5- to 2.0-fold lower than that of omalizumab. That is, it was found that a binding affinity thereof to substances other than IgE is 1.5- to 2.0-fold lower than that of omalizumab. In addition, the dissociation rate (kd) value of the polypeptide dimer according to an embodiment of the present invention was measured to be 40- to 106-fold higher than that of omalizumab. As a result, it was confirmed that the polypeptide dimer according to an embodiment of the present invention has an equilibrium dissociation constant (KD<kd/ka>) value which is 22- to 69-fold higher than omalizumab. From this, it was identified that the polypeptide dimer according to an embodiment of the present invention has a remarkably increased binding affinity to IgE as compared with omalizumab. In particular, it was identified that the polypeptide dimer to which sialic acid is added (FcεRIαECD-Fc2ST) exhibits the highest IgE-binding affinity which is 69-fold higher than omalizumab.

In addition, as shown in FIGS. 3 and 4, it was confirmed that omalizumab loses its binding to IgE when a certain period of time has passed after the binding, whereas the polypeptide dimer according to an embodiment of the present invention (FcεRIαECD-Fc2ST, IgE_{TRAP}) is not separated from IgE once it binds to IgE. That is, it was confirmed that the polypeptide dimer according to an embodiment of the present invention is not easily separated from IgE, and has a much better ability to maintain its bound state than omalizumab.

Experimental Example 3. Confirmation of Binding Affinity of Polypeptide Dimer to IgG Receptors

The binding affinity of a polypeptide dimer according to an embodiment of the present invention (FcεRIαECD-Fc2ST, IgE_{TRAP}) or omalizumab (Xolair) to Fc gamma receptors of IgG was confirmed using Octet RED384 system (Pall ForteBio, CA, USA).

Specifically, FcγRI, FcγRIIA, FcγRIIB, FcγRIIIA, and FcγRIIIB recombinant proteins (R & D Systems Inc., 5 μg/ml) which are Fc gamma receptors were immobilized in 300 mM acetate buffer (pH 5) on the activated AR2G biosensor. As a running buffer, PBS containing 0.1% Tween-20 and 1% bovine serum was used. All measurements were carried out at 30° C. with a sample plate shaker at a rate of 1,000 rpm. The results are shown in FIGS. 5a to 5e, and binding affinity of omalizumab and the polypeptide dimer to the IgG Fc gamma receptors are quantified and shown in FIG. 6.

As a result, it was confirmed that omalizumab showed a high binding affinity to Fc gamma receptors of IgG, whereas the polypeptide dimer had a significantly low binding affinity to Fc gamma receptors of IgG. Through this, it was confirmed that the polypeptide dimer does not bind with Fc gamma receptors of IgG.

Experimental Example 4. Confirmation of Activity of Polypeptide Dimer Through Beta-Hexosaminidase Assay in Mouse Bone Marrow-Derived Mast Cells

Beta-hexosaminidase assay was performed for in vitro activity analysis of the polypeptide dimer according to an embodiment of the present invention.

Specifically, the polypeptide dimer (FcεRIαECD-Fc2) according to an embodiment of the present invention was mixed, at each concentration, with IgE (1 μg/mL), and incubated at 20° C. for 30 minutes to prepare samples. Mouse bone marrow-derived mast cells in culture for mast cell activation were washed with Hank's balanced salt solution (HBSS) buffer to remove the medium, the number of cells was measured, and then 5×10^5 cells were refloated into 40 μL of HBSS buffer.

50 μL of the sample thus prepared was added to activated mast cells. Then, the resultant was incubated in a 5% CO₂ incubator at 37° C. for 30 minutes. Subsequently, after the addition of each 10 μL of DNP (2,4-dinitrophenol, 100 ng/mL), which is a foreign antigen, incubation was performed again at 37° C. for 30 minutes in 5% CO₂, and then 30 μL of the supernatant was separated.

30 μL of the separated supernatant and 30 μL of the substrate (4-nitrophenyl N-acetyl-β-D-glucosaminide, 5.84 mM) were mixed well, and then incubated at 37° C. for 20 minutes in 5% CO₂. Then, 140 μL of 0.1 M sodium carbonate buffer (pH 10) as a stop solution was added to terminate the reaction. Thereafter, absorbance at 405 nm was measured to identify a secretion amount of β-hexosaminidase secreted by the foreign antigen in the activated mast cells. The results are shown in FIG. 7.

As shown in FIG. 7, it was confirmed that the activity of IgE is considerably inhibited from the case in which the concentration of IgE and the polypeptide dimer of the present invention is 1:1. That is, it was confirmed that the polypeptide dimer according to an embodiment of the present invention reacts even when the concentration thereof is the same as that of IgE.

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Experimental Example 5. Comparison of Activity
of Polypeptide Dimer and Human Anti-IgE
Antibody Using β -Hexosaminidase Assay in
Human Fc ϵ RI-Expressing Mouse Bone
Marrow-Derived Mast Cells

In order to compare the activity of the polypeptide dimer according to an embodiment of the present invention (Fc ϵ RI α ECD-Fc2ST, IgE_{TRAP}) and Xolair (omalizumab), which is a human anti-IgE antibody, a beta-hexosaminidase analysis was performed.

The respective polypeptide dimers and Xolair were prepared at each concentration, and then mixed with human IgE (1 μ g/mL). Then, incubation was performed at room temperature for 30 minutes to prepare samples. In addition, mast cells derived from and differentiated from mouse bone marrow, in which a human Fc ϵ RI gene had been introduced and the mouse Fc ϵ RI gene had been removed, were prepared. The prepared mast cells were washed with HBSS buffer, and then 5×10^5 cells were refloated into 60 μ L of HBSS buffer.

Then, 20 μ L of the sample thus prepared was added to the mast cells, and then incubated in a 5% CO₂ incubator at 37° C. for 30 minutes. Subsequently, after 20 μ L of human anti-IgE antibody (BioLegend, Cat No. 325502, 0.5 μ g/mL) was added, the resultant was incubated again in 5% CO₂ incubator at 37° C. for 30 minutes. Then, after centrifugation at 1,500 rpm at 4° C., 30 μ L of the supernatant was separated. 30 μ L of the separated supernatant and 30 μ L of the substrate (4-nitrophenyl N-acetyl- β -glucosaminide, 5.84 mM) were mixed well, and then incubated in a 5% CO₂ incubator at 37° C. for 25 minutes. Then, 140 μ L of 0.1 M sodium carbonate buffer (pH 10) was added to terminate the reaction. Thereafter, absorbance at 405 nm was measured to compare relative amounts of secreted β -hexosaminidase, and an inhibitory effect against mast cells was identified depending on each concentration of the sample. The results are shown in FIG. 8.

As shown in FIG. 8, IC₅₀ of the polypeptide dimer was measured to be approximately 11.16 ng/mL, and IC₅₀ of Xolair was measured to be approximately 649.8 ng/mL. Therefore, it was identified that the polypeptide dimer has a 58-fold higher inhibitory ability on mast cell activity than Xolair.

Experimental Example 6. Confirmation of Activity
of Polypeptide Dimer Through In Vivo Assay in
Food Allergy Model

50 μ g of ovalbumin (OVA) and 1 mg of alum were intraperitoneally administered to Balb/c mice (Orientbio Inc.) two times at a 14-day interval to induce sensitization. Thereafter, 50 mg of OVA was orally administered five times in total on days 28, 30, 32, 34, and 36, to induce food allergy in intestines.

After the OVA was orally administered two times, that is, on day 31, the mice were divided into three groups, each containing 7 mice. The three divided groups were as follows: Group 1 receiving the polypeptide dimer (Fc ϵ RI α ECD-Fc2ST) at a high concentration (200 μ g),

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Group 2 receiving the polypeptide dimer (Fc ϵ RI α ECD-Fc2ST) at a low concentration (20 μ g), and Group 3 receiving nothing.

While orally administering the OVA, it was observed whether diarrhea occurs due to food allergy induction. In addition, the mice were sacrificed on day 37, and the number of mast cells in the small intestine, the IgE concentration in blood, and the concentration of the mast cell degranulation enzyme (mast cell protease-1 (MCPT-1)) in blood were analyzed for the mice belonging to each group.

As a result, as shown in FIG. 9, diarrhea occurred in mice of Group 3 after the second oral administration of OVA. On the other hand, diarrhea occurred in mice of Group 1 and Group 2 after the third oral administration of OVA. In particular, the frequency of diarrhea in mice of Group 1 was lower than that in mice of Group 2. Through this, it was confirmed that the effect against food allergy increased in proportion to the concentration of a polypeptide dimer.

Experimental Example 7. Analysis of Content of
Sialic Acid of Polypeptide Dimer

Based on the aspects that the production rate of the polypeptide produced from the Fc ϵ RI α ECD-Fc2+a2,6-ST cell line prepared in Example 1 above is the highest, and the anti-allergic effect of the polypeptide dimer (Fc ϵ RI α ECD-Fc2ST) in Experimental Examples 4 to 6 is excellent, the sialic acid content of the polypeptide dimer was analyzed to investigate the anti-allergic efficacy of the polypeptide dimer depending on the sialic acid content.

Specifically, in order to measure the sialic acid content contained in the glycan structure of the polypeptide dimer (Fc ϵ RI α ECD-Fc2ST) produced from the Fc ϵ RI α ECD-Fc2+a2,6-ST cell line prepared in Example 1 above, sialic acid was first separated by treatment with sialidase, which is an enzyme associated with sialic acid catabolism. Thereafter, the separated sialic acid was isolated, detected and quantified using HPLC (waters, alliance e2659).

The polypeptide dimer was divided into three samples according to the pH gradient, and isolated and purified. Three samples were placed in an AmiconUltra 10K (Millipore, UFC501096) filter, and centrifuged for 10 minutes under a condition of 13,000 rpm, 4° C. temperature, and repeated 5 times, and the concentrate was exchanged with deionized water and concentrated. The concentration of the sample was set to 10 mg/mL or more when measured at a wavelength of 280 nm.

Thereafter, 0.5 mg of each sample was taken and placed in an EP tube, and 1 μ L of sialidase (Roche, 10 269 611 001) and 40 μ L of sodium phosphate buffer (pH 7.0) at a concentration of 10 mM were added, and then deionized water was added so that the final volume was 100 μ L. 2 μ L of each sample was taken to measure the concentration at a wavelength of 280 nm, and the confirmed concentration value was used as the final analysis concentration.

The sample was reacted for 18 hours in a 37° C. incubator, and then placed in an AmiconUltra 10K filter, and centrifuged for 15 minutes under a condition of 13,000 rpm, 4° C., and the filtrate passing through a filter was used for analysis. HPLC analysis conditions are shown in Table 4 below.

TABLE 4

Analysis Column	RHM-monosaccharide H+ (8%) 300 $\times$ 7.8 mm (Rezex); analysis column
Standard Substance	RHM-monosaccharide H+ (8%) 50 $\times$ 7.8 mm (Rezex); guard column
Flux	NGNA: 2~40 μ M NANA: 100~2000 μ M 0.55 ml/min

TABLE 4-continued

Column Temperature	50° C.
Detection	206 nm
Injection Volume	5 µL
Mobile Phase	5 mM sulfuric acid in water
Gradient/Time	Isocratic Elution/45 minutes

The standard substance for calculating the content of sialic acid is a mixture of N-acetylneuraminic acid (hereinafter referred to as NANA) and N-glycolylneuraminic acid (hereinafter referred to as NGNA). The linear regression equation for the standard curve was obtained to calculate the molar concentrations of NGNA and NANA in the analyte. The mixtures used are shown in Table 5 below.

TABLE 5

Mixture	1	2	3	4	5
NANA (µM)	2,000	1,000	500	300	200
NGNA (µM)	40	20	10	6	4

The NANA sialic acid content in the sample was calculated using the NANA content. The sialic acid content in the sample was expressed as a molar ratio of sialic acid/sample (mol/mol).

$$\text{Sialic acid content of Sample} = \frac{(\text{molar concentration of NANA in sample})}{(\text{Molar concentration of sample})} \quad [\text{Equation 1}]$$

In addition, the content of NGNA in the sample was calculated using the NGNA content. The content of NGNA in the sample was expressed as a molar ratio of NGNA/analyte (mol/mol).

$$\text{NGNA Content of sample} = \frac{(\text{NGNA molar concentration in analyte})}{(\text{Molar concentration of analyte})} \quad [\text{Equation 2}]$$

The results are shown in Table 6 below.

TABLE 6

Sample		NGNA/Protein (mol/mol)	NANA/Protein (mol/mol)
FcεRIαECD-Fc2ST	Sample 1 (SA ^{low})	0.13	7.7
	Sample 2 (SA ^{medi})	0.17	12.0
	Sample 3 (SA ^{high})	0.27	19.1

As shown in Table 6, it was divided into three samples according to the pH gradient, and separated and purified, and the sialic acid content in the samples was measured differently. In order to compare the anti-allergic efficacy according to the sialic acid content, a polypeptide dimer is referred to as "SA^{low}", "SA^{medi}" and "SA^{high}" in order of the sialic acid content.

Experimental Example 8. Confirmation of Change in Dimer Formation of Polypeptide According to Sialic Acid Content

In order to confirm whether there was a change in dimer formation of the polypeptide according to the sialic acid content, SA^{low}, SA^{medi} and SA^{high} separated in Example 9 were subjected to SDS-PAGE analysis under a non-reducing condition and a reducing condition in the same manner as in Experimental Example 1.

As a result, in all SA^{low}, SA^{medi} and SA^{high}, a polypeptide having a size of about 150 kDa was detected under a non-reducing condition, and a polypeptide having a size of about 75 kDa was detected under a reducing condition. Through this, it was confirmed that the polypeptide forms a dimer (FIG. 10).

Experimental Example 9. Confirmation of Isoelectric Point of Polypeptide Dimer According to Sialic Acid Content

In order to confirm whether the isoelectric point of the polypeptide dimer changes according to the sialic acid content, SA^{low}, SA^{medi} and SA^{high} Separated in Experimental Example 7 were mixed with the IEF sample solution, and then loaded into a pH 3-7 IEF gel (Invitrogen), and electrophoresis was sequentially performed for 1 hour under a 100V condition, for 1 hour under a 200V condition, and for 30 minutes under a 500V condition. After the electrophoresis was completed, the gel was reacted in a fixing solution comprising 12% trichloroacetic acid and 3.5% sulfosalicylic acid for 30 minutes, and then washed with distilled water. A protein was stained with a Coomassie Brilliant Blue solution.

As a result, the isoelectric points of SA^{low}, SA^{medi} and SA^{high} were slightly different according to the sialic acid content, and were found to be about 5.3, 4.9 and 4.7, respectively (FIG. 11).

Experimental Example 10. Confirmation of Anti-Allergic Activity of Polypeptide Dimer According to Sialic Acid Content in Food Allergy Animal Model: Measurement of Frequency of Diarrhea

In order to confirm whether the activity of the polypeptide dimer changes according to the sialic acid content, sensitization was induced by intraperitoneal administration of 50 µg of OVA and 1 mg of alum twice at an interval of 14 days for Balb/c mice (Orient Bio). Thereafter, at Day 28, 30, 32, 34, 36, 38, 40 over a total of 7 times, food allergy was induced by oral administration of 50 mg of OVA. In this case, the oral administration of OVA was performed after fasting for 4 hours.

The OVA was administered orally twice, and then at Day 40, the mice were divided into 3 groups of 7 mice. The mice were divided into Group 1, a group in which a polypeptide dimer having a high content of sialic acid (SA^{high}) is subcutaneously administered at a high concentration (200

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μg), Group 2, a group in which a polypeptide dimer having a low content of sialic acid (SA^{low}) is subcutaneously administered at a high concentration (200 μg), and Group 3, a group in which PBS is subcutaneously administered. It was confirmed whether diarrhea occurred according to the induction of food allergy when the OVA was administered orally.

As a result, diarrhea occurred in mice of Group 3 after the fourth oral administration of OVA. Diarrhea occurred in mice of Group 2 after the fifth oral administration of OVA. In particular, the frequency of diarrhea increased rapidly in mice of Group 2 after the sixth oral administration of OVA. On the other hand, diarrhea occurred in mice of Group 1 after the sixth oral administration of OVA, but the frequency of diarrhea was maintained but did not increase in mice of Group 1 even after the seventh oral administration of OVA (FIG. 12). Through this, it was confirmed that the polypeptide dimer having a high content of sialic acid (SA^{high}) had an excellent anti-allergic effect compared to the polypeptide dimer having a low content of sialic acid (SA^{low}).

Experimental Example 11. Confirmation of
Anti-Allergic Activity of Polypeptide Dimer
According to Sialic Acid Content in Food Allergy
Animal Model: Measurement of IgE Concentration
in Blood

The blood was collected by orbital sampling method from mice of each group in which the seventh oral administration of OVA of Experimental Example 10 was performed. After reacting for 30 minutes at ambient temperature, serum was separated by centrifugation for 15 minutes under a condition of 13,000 rpm at a temperature of 4° C. In order to measure free IgE concentration in the blood, mouse total IgE ELISA kit (BioLegend) was used, and in this case, ELISA was performed according to the manufacturer's protocol except that a polypeptide dimer (SA^{low} and SA^{high}) according to an embodiment of the present invention was used as a substance coated on a 96-well-plate instead of an anti-IgE antibody.

Specifically, a 96-well-plate was coated with a polypeptide dimer (SA^{low} and SA^{high}) diluted with PBS, and reacted overnight at a temperature of 4° C. The next day, after washing with PBS containing 0.05% tween 20 (hereinafter, washing buffer), a blocking buffer (assay diluent) was added and reacted for 1 hour. Thereafter, it was washed with a washing buffer, and then the mouse IgE to be used as a standard solution and the serum samples of mice were diluted in a 1× assay diluent, and placed in a plate and reacted for 2 hours.

After washing again with a washing buffer, biotin-labeled mouse anti-IgE antibody was added and reacted for 1 hour. After washing with a washing buffer, HRP (horse radish peroxidase)-labeled avidin (Avidin-HRP) was added and reacted for 30 minutes. After washing with a washing buffer, a substrate solution was added and reacted for 20 minutes while blocking light, and then a stop solution (1 M H₂SO₄) was added to stop the reaction. Thereafter, the absorbance value was measured at a wavelength of 450 nm with a microplate reader (Epoch Microplate Spectrophotometer), and the concentration was calculated.

As a result, for the mice of Group 3 in which PBS was subcutaneously administered, the IgE concentration in the blood was calculated to be about 8,000 ng/mL. In addition, for the mice of Group 2 in which a polypeptide dimer having a low content of sialic acid (SA^{low}) was subcutaneously administered, the IgE concentration in the blood was calculated to be about 7,000 ng/mL. On the other hand, for the

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mice of Group 1 in which a polypeptide dimer having a high content of sialic acid (SA^{high}) was subcutaneously administered, the IgE concentration in the blood was calculated to be about 4,900 ng/mL, and there was a significant difference from the IgE concentration values in the blood between mice of Group 2 and Group 3 (FIG. 13).

From these results, it was found that when the polypeptide dimer having a high content of sialic acid was administered subcutaneously, the IgE content in the blood was reduced. In addition, based on such reduction of IgE in the blood, it was predicted that the polypeptide dimer having a high content of sialic acid is effective in treating allergy.

Experimental Example 12. Confirmation of
Anti-Allergic Activity of Polypeptide Dimer
According to Sialic Acid Content in Food Allergy
Animal Model: Measurement of MCPT-1
Concentration in Blood

Serum was prepared by taking blood from mice of each group 2 days after the food allergy experiment was completed. In order to measure the MCPT-1 (Mast Cell Protease-1) concentration in the blood, an MCPT-1 ELISA kit (Invitrogen) was used to perform the procedure according to the manufacturer's protocol.

Specifically, a 96-well-immunized plate was coated with a mouse anti-MCPT-1 antibody and reacted overnight at a temperature of 4° C. The next day, after washing with PBS containing 0.05% tween 20 (hereinafter, washing buffer), PBS containing 1% BSA (bovine serum albumin) (hereinafter, blocking buffer, assay diluent) was added and reacted for 1 hour. Thereafter, it was washed with a washing buffer, and then the MCPT-1 to be used as a standard solution and the serum samples of mice were diluted in a 1× assay diluent, and placed in a plate and reacted for 2 hours.

After 2 hours, a biotin-labeled mouse anti-MCPT-1 antibody was added and reacted for 1 hour. After washing with a washing buffer, HRP (horse radish peroxidase)-labeled avidin (Avidin-HRP) was added and reacted for 30 minutes. After washing with a washing buffer, a substrate solution was added and reacted for 20 minutes while blocking light, and then a stop solution (1 M H₂SO₄) was added to stop the reaction. Thereafter, the absorbance value excluding that measured at a wavelength of 570 nm wavelength from that measured at a wavelength of 450 nm was determined with a microplate reader (Epoch Microplate Spectrophotometer) and the concentration was calculated.

As a result, for the mice of Group 3 in which PBS was subcutaneously administered, the MCPT-1 concentration in the blood was calculated to be about 4,000 ng/mL. In addition, for the mice of Group 2 in which a polypeptide dimer having a low content of sialic acid (SA^{low}) was subcutaneously administered, the MCPT-1 concentration in the blood was calculated to be about 4,200 ng/mL. On the other hand, for the mice of Group 1 in which a polypeptide dimer having a high content of sialic acid (SA^{high}) was subcutaneously administered, the MCPT-1 concentration in the blood was calculated to be about 2,800 ng/mL, and there was a significant difference from the MCPT-1 concentration values in the blood between mice of Group 2 and Group 3 (FIG. 14).

Example 3. Preparation of Polypeptide Dimer
Having High Content of Sialic Acid Through
Purification

In order to confirm the physical properties of a polypeptide dimer according to the sialic acid content, a polypeptide

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dimer containing a wide range of sialic acid was obtained. Specifically, a polypeptide dimer obtained from the FcεRIαECD-Fc2+a2,6-ST cell line was purified using affinity chromatography and anion exchange chromatography.

First, the cell culture solution was subjected to affinity chromatography to remove the primary impurities from the culture solution. The affinity chromatography was performed using Amsphere™ A3 pre-packing column (bed height 5 cm, volume 5 mL) packed with Amsphere™ A3 resin and AKTA Avant 25 equipment, a liquid chromatography system. Here, it was performed using 100 mM Glycine, pH 3.3 as an eluted buffer.

Thereafter, the anion exchange chromatography was performed using Hiscreen QFF (bed height 10 cm, volume 5 mL) packed with Q Sepharose Fast Flow resin and AKTA Avant 25 equipment, a liquid chromatography system. 20 mM phospho-citrate, pH 3.5 was used as an eluted buffer. In this case, the concentration of 20 mM phospho-citrate, pH 4.0, as a washing buffer, was adjusted to obtain dimers having various sialic acid contents. Analysis of the sialic acid content was performed by the analysis method used in Experimental Example 7. Through this, polypeptide dimers having a sialic acid content of 7.0 mol/mol, 10.3 mol/mol, 12.9 mol/mol, 14.9 mol/mol and 21.4 mol/mol were obtained.

Experimental Example 13. Pharmacokinetics Analysis of Polypeptide Dimer According to Sialic Acid Content in Mouse Model

In order to perform pharmacokinetics analysis according to the sialic acid content of the polypeptide dimer, the polypeptide dimers having a sialic acid content of 7.0 mol/mol, 10.3 mol/mol, 12.9 mol/mol, 14.9 mol/mol and 21.4 mol/mol prepared in Example 3 was subcutaneously injected into mice at a concentration of 10 mg/kg, respectively. In order to analyze the pharmacokinetics of the polypeptide dimer, the blood was collected at Hour 0, Hour 3, Hour 10, Hour 24, Hour 72, Hour 168, and Hour 240 after administration of the substance and analyzed in the following manner.

The anti-FcεRI antibody (Invitrogen) was diluted in 1×PBS (Welgene) at a concentration of 0.5 μg/mL, and added to an immunoplate (Thermo), and reacted overnight under a condition of 4° C. After the reaction was completed, the immunoplate was washed with 1×PBST, and I-Block solution (Invitrogen), a blocking solution, was added per well and reacted at ambient temperature for 1 hour. During blocking, standard and analytical samples were prepared. The polypeptide dimer was diluted in 1% BSA/PBS containing mouse blank serum at a concentration of 0.19 ng/mL to 200 ng/mL to prepare a standard substance, and the dilution with 1/20-1/300 using 1% BSA/PBS containing the blank serum was also performed to prepare an analytical sample.

After the blocking was completed, the plate was washed with 1×PBST, and the prepared standard and analytical samples were added per well, and then reacted at ambient temperature for 1 hour. Thereafter, it was washed with 1×PBST, and an anti-human IgG4-HRP (Southern biotech) diluted to 1:10,000 was added, and then reacted at ambient temperature for 1 hour. After the reaction was completed, the plate was washed with 1×PBST, and TMB solution (Thermo) was added to the well and reacted. When the absorbance value was measured in the range of 0.8-1.0 at a wavelength of 650 nm using a microplate reader (Molecular devices), a stop solution (Sigma) was added per well to stop

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the reaction. After the reaction was terminated, the absorbance was measured in the plate at a wavelength of 450 nm within 5 minutes and the numerical values were analyzed. The resulting pharmacokinetic graph and pharmacokinetic parameters of the polypeptide dimer according to the sialic acid contents are shown in FIG. 15 and Table 7.

As a result, as shown in Table 7, it was confirmed that the higher the sialic acid content of the polypeptide dimer administered to the mouse, the higher the concentration in the blood.

TABLE 7

sialic acid content (mol/mol)	T _{1/2} (hr)	T _{max} (hr)	C _{max} (μg/ml)	AUC _{last} (hr*μg/ml)
7.0	51.3 ± 5.9	5.3 ± 3.3	1.3 ± 0.5	47.8 ± 3.2
10.3	44.5 ± 6.7	5.3 ± 3.3	1.6 ± 0.6	57.1 ± 9.6
12.9	40.0 ± 9.5	17 ± 9.9	2.0 ± 0.4	119.7 ± 5.2
14.9	35.4 ± 3.5	7.7 ± 3.3	2.6 ± 0.3	182.7 ± 7.0
21.4	38.5 ± 5.0	10.0 ± 0.0	12.1 ± 0.9	890.0 ± 64.4

Experimental Example 14. Confirmation of Efficacy According to Sialic Acid Content of Polypeptide Dimer in Passive Systemic Anaphylaxis Mouse Model

In order to confirm the difference in an anti-allergic effect in vivo according to the sialic acid content of the polypeptide dimer, the method disclosed in the literature was applied to induce the passive systemic anaphylaxis (PSA) in mice in the manner as follows (Methods and Protocols, Methods in Molecular Biology, vol. 1032, DOI 10.1007/978-1-62703-496-8_10). It is known that mast cells are activated by induction of anaphylaxis and body temperature is reduced by increasing mast cells. While inducing PSA in mice using this, the polypeptide dimer having the respective sialic acid content was administered, and then the body temperature was measured to compare the anti-allergic effect.

The body temperature was measured in transponder-inserted Balb/c mice (Orient Bio), and then 20 μg of anti-DNP IgE (Sigma) containing 0.9% NaCl was intraperitoneally administered, and then the polypeptide dimers having different sialic acid contents, respectively, were subcutaneously administered at 10 mg/kg. After 24 hours, the body temperature of the mice was measured before DNP-HAS (Sigma) containing 0.9% NaCl, a foreign antigen, was administered. Thereafter, 1 mg of the antigen was administered intravenously. After administration, the body temperature of the mice was measured at an interval of 10 minutes for 1 hour using a remote body temperature meter (Bio medic data systems). The experimental results are shown in FIG. 16.

As a result, it was confirmed that the body temperature of the mice was reduced in groups in which a vehicle and a polypeptide dimer having a sialic acid content of 10.3 mol/mol were administered, and the body temperature of the mice was not reduced in groups in which the polypeptide dimer having a sialic acid content of 14.9 mol/mol or more was administered. Through this, it was confirmed that a polypeptide dimer having a high content of sialic acid has an excellent effect of reducing the induction of passive systemic anaphylaxis compared to a polypeptide dimer having a low content of sialic acid.

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50           55           60
Glu Tyr Lys Cys Gln His Gln Gln Val Asn Glu Ser Glu Pro Val Tyr
65           70           75           80
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115          120          125
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130          135          140
Asp Ser Gly Thr Tyr Tyr Cys Thr Gly Lys Val Trp Gln Leu Asp Tyr
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20           25           30
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35           40           45
Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe
50           55           60
Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp
65           70           75           80
Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu
85           90           95
Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg
100          105          110

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Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met Thr Lys
 115 120 125

Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp
 130 135 140

Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys
 145 150 155 160

Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser
 165 170 175

Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val Phe Ser
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cacaatggca gcctgagcga ggagaccaac agctccctga acatcgtgaa cgccaagttc 180

gaggacagcg gcgagtacaa gtgccagcac cagcaggtga acgagagcga gcccggttac 240

ctggaggtgt tcagcgactg gctgctgctg caggccagcg ccgaggtggt gatggagggc 300

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 35 40 45

Leu Thr Cys Asn Gly Asn Asn Phe Phe Glu Val Ser Ser Thr Lys Trp
 50 55 60

Phe His Asn Gly Ser Leu Ser Glu Glu Thr Asn Ser Ser Leu Asn Ile
 65 70 75 80

Val Asn Ala Lys Phe Glu Asp Ser Gly Glu Tyr Lys Cys Gln His Gln
 85 90 95

Gln Val Asn Glu Ser Glu Pro Val Tyr Leu Glu Val Phe Ser Asp Trp
 100 105 110

Leu Leu Leu Gln Ala Ser Ala Glu Val Val Met Glu Gly Gln Pro Leu
 115 120 125

Phe Leu Arg Cys His Gly Trp Arg Asn Trp Asp Val Tyr Lys Val Ile
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Tyr Tyr Lys Asp Gly Glu Ala Leu Lys Tyr Trp Tyr Glu Asn His Asn
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Ile Ser Ile Thr Asn Ala Thr Val Glu Asp Ser Gly Thr Tyr Tyr Cys
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Thr Gly Lys Val Trp Gln Leu Asp Tyr Glu Ser Glu Pro Leu Asn Ile
 180 185 190

Thr Val Ile Lys Ala Pro Arg Glu Lys Tyr Trp Leu Gln Arg Asn Thr
 195 200 205

Gly Arg Gly Gly Glu Glu Lys Lys Gly Ser Lys Glu Lys Glu Glu Gln
 210 215 220

Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro Ser His Thr Gln Pro
 225 230 235 240

Leu Gly Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr Leu Met Ile
 245 250 255

-continued

Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val Ser Gln Glu
 260 265 270
 Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val Glu Val His
 275 280 285
 Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg
 290 295 300
 Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu Asn Gly Lys
 305 310 315 320
 Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu
 325 330 335
 Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr
 340 345 350
 Thr Leu Pro Pro Ser Gln Glu Glu Met Thr Lys Asn Gln Val Ser Leu
 355 360 365
 Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp
 370 375 380
 Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val
 385 390 395 400
 Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp
 405 410 415
 Lys Ser Arg Trp Gln Glu Gly Asn Val Phe Ser Cys Ser Val Met His
 420 425 430
 Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu
 435 440 445
 Gly Lys
 450

<210> SEQ ID NO 12
 <211> LENGTH: 1350
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: nucleotides sequence of FcεRIα ECD-hinge-Fc2
 <400> SEQUENCE: 12

atggacgcc	tgctgagagg	cctgtgctgt	gtgctgctgc	tgtagcgcg	cgtgttcgtg	60
tcccctagcc	acgccgtgcc	ccagaagccc	aaggtgagcc	tgaaccctcc	ctggaacaga	120
atcttcaagg	gcgagaacgt	gacctgacc	tgcaacggca	acaacttctt	cgaggtgagc	180
agcaccaagt	ggttccacaa	tggcagcctg	agcgaggaga	ccaacagctc	cctgaacatc	240
gtgaacgcc	agttcgagga	cagcggcgag	tacaagtgcc	agcaccagca	ggtgaacgag	300
agcgagccc	gtacctgga	ggtgttcagc	gactggctgc	tgctgcaggc	cagcgccgag	360
gtggtgatgg	agggccagcc	cctgttcctg	agatgccacg	gctggagaaa	ctgggacgtg	420
tacaaggtga	tctactacaa	ggatggcgag	gccctgaagt	actggtacga	gaaccacaac	480
atctccatca	ccaacgccac	cgtggaggac	agcggcacct	actactgcac	aggcaaggtg	540
tggcagctgg	actacgagag	cgagcccctg	aacatcaccg	tgatcaaggc	tcccagagag	600
aagtactggc	tgcaaggaga	caccggcaga	ggaggcgagg	aaaagaaagg	aagcaaggag	660
aaggaggagc	aggaggaaag	agaaccaag	acccccaggt	gccccagcca	caccagcccc	720
ctgggcgtgt	tcctgttccc	ccccaaagcc	aaggacaccc	tgatgatcag	cagaaccccc	780
gaggtgacct	gcgtggtcgt	ggatgtgagc	caggaagatc	ccgaagtgca	gttcaactgg	840
tacgtggatg	gcgtggaagt	gcacaacgcc	aagaccaagc	ccagagaaga	gcagttcaac	900

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tccacctaca gagtggtag cgtgctgacc gtgctgcacc aggactggct gaacggcaag   960
gagtacaagt gcaaggtgtc caacaaaggc ctgcccagct ccatcgagaa gaccatcagc  1020
aaagccaaag gccagcccag agaaccacag gtgtacaccc tgcctcccag ccaggaagag  1080
atgaccaaga accaggtgtc cctgacctgc ctggtgaaag gcttctaccc cagcgacatc  1140
gccgtggagt gggaaagcaa cgccagccc gagaaacaatt acaagacaac ccctcccggtg  1200
ctggatagcg atggcagctt ctttctgtac agcagactga ccgtggacaa gagcagatgg  1260
caggaaggca acgtgttcag ctgcagcgtg atgcacgaag ccctgcacaa ccactacacc  1320
cagaagagcc tgtccctgag cctgggcaag                                     1350

```

```

<210> SEQ ID NO 13
<211> LENGTH: 469
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: FceRIa ECD-hinge-Fc3

```

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<400> SEQUENCE: 13

```

```

Met Asp Ala Met Leu Arg Gly Leu Cys Cys Val Leu Leu Leu Cys Gly
1           5           10          15

Ala Val Phe Val Ser Pro Ser His Ala Val Pro Gln Lys Pro Lys Val
20          25          30

Ser Leu Asn Pro Pro Trp Asn Arg Ile Phe Lys Gly Glu Asn Val Thr
35          40          45

Leu Thr Cys Asn Gly Asn Asn Phe Phe Glu Val Ser Ser Thr Lys Trp
50          55          60

Phe His Asn Gly Ser Leu Ser Glu Glu Thr Asn Ser Ser Leu Asn Ile
65          70          75          80

Val Asn Ala Lys Phe Glu Asp Ser Gly Glu Tyr Lys Cys Gln His Gln
85          90          95

Gln Val Asn Glu Ser Glu Pro Val Tyr Leu Glu Val Phe Ser Asp Trp
100         105         110

Leu Leu Leu Gln Ala Ser Ala Glu Val Val Met Glu Gly Gln Pro Leu
115         120         125

Phe Leu Arg Cys His Gly Trp Arg Asn Trp Asp Val Tyr Lys Val Ile
130         135         140

Tyr Tyr Lys Asp Gly Glu Ala Leu Lys Tyr Trp Tyr Glu Asn His Asn
145         150         155         160

Ile Ser Ile Thr Asn Ala Thr Val Glu Asp Ser Gly Thr Tyr Tyr Cys
165         170         175

Thr Gly Lys Val Trp Gln Leu Asp Tyr Glu Ser Glu Pro Leu Asn Ile
180         185         190

Thr Val Ile Lys Ala Pro Arg Glu Lys Tyr Trp Leu Gln Ala Gln Pro
195         200         205

Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr Ala Pro Ala Thr Thr
210         215         220

Arg Asn Thr Gly Arg Gly Glu Glu Lys Lys Gly Ser Lys Glu Lys
225         230         235         240

Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro Ser His
245         250         255

Thr Gln Pro Leu Gly Val Phe Leu Phe Pro Pro Lys Pro Lys Asp Thr
260         265         270

Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val Val Val Asp Val

```

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275	280	285
Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr Val Asp Gly Val 290 295 300		
Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu Gln Phe Asn Ser 305 310 315 320		
Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His Gln Asp Trp Leu 325 330 335		
Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys Gly Leu Pro Ser 340 345 350		
Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln Pro Arg Glu Pro 355 360 365		
Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met Thr Lys Asn Gln 370 375 380		
Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro Ser Asp Ile Ala 385 390 395 400		
Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn Tyr Lys Thr Thr 405 410 415		
Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu Tyr Ser Arg Leu 420 425 430		
Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val Phe Ser Cys Ser 435 440 445		
Val Met His Glu Ala Leu His Asn His Tyr Thr Gln Lys Ser Leu Ser 450 455 460		
Leu Ser Leu Gly Lys 465		

<210> SEQ ID NO 14

<211> LENGTH: 1407

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: nucleotides sequence of FceRIa ECD-hinge-Fc3

<400> SEQUENCE: 14

```

atggaaccca tgctgagagg cctgtgctgt gtgctgctgc tgtgcggcgc cgtgttcgtg    60
tcccctagcc acgccgtgcc ccagaagccc aaggtagacc tgaaccctcc ctggaacaga    120
atcttcaagg gcgagaacgt gacctgacc tgcaacggca acaacttctt cgaggtgagc    180
agcaccaagt ggttcacaaa tggcagcctg agcgaggaga ccaacagctc cctgaacatc    240
gtgaacgcca agttcgagga cagcggcgag tacaagtgcc agcaccagca ggtgaacgag    300
agcgagcccc tgtacctgga ggtgttcagc gactggctgc tgctgcaggc cagcgccgag    360
gtggtgatgg agggccagcc cctgttcttg agatgccacg gctggagaaa ctgggacgtg    420
tacaagtgta tctactacaa ggatggcgag gccctgaagt actggtacga gaaccacaac    480
atctccatca ccaacgccac cgtggaggac agcggcacct actactgcac aggcaaggtg    540
tggcagctgg actacgagag cgagcccttg aacatcaccg tgatcaaggc tcccagagag    600
aagtactggc tgcaggccca gccccaggcc gagggcagcc tggctaaggc caccacagct    660
cccgccacca ccaggaacac cggcagagga ggcgaggaaa agaaaggaag caaggagaag    720
gaggagcagg aggaagagaa aaccaagacc cccgagtgcc ccagccacac ccagcccctg    780
ggcgtgttcc tgttccccc caagcccaag gacacctga tgatcagcag aacccccgag    840
gtgacctgcy tggctgtgga tgtgagccag gaagatcccg aagtgcagtt caactggtac    900
gtggatggcg tggaagtga caacgccaa accaagccca gagaagagca gttcaactcc    960

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acctacagag tggtagcggt gctgaccgtg ctgcaccagg actgggtgaa cggcaaggag 1020
tacaagtgca aggtgtccaa caaaggcctg cccagctcca tcgagaagac catcagcaaa 1080
gccaaaggcc agcccagaga accccaggtg tacaccctgc ctcccagcca ggaagagatg 1140
accaagaacc aggtgtccct gacctgctg gtgaaaggct tctaccccag cgacatcgcc 1200
gtggagtggg aaagcaacgg ccagcccag aacaattaca agacaacccc tcccgctgctg 1260
gatagcgatg gcagcttctt tctgtacagc agactgaccg tggacaagag cagatggcag 1320
gaaggcaacg tgttcagctg cagcgtgatg cacgaagccc tgcacaacca ctacaccag 1380
aagagcctgt ccctgagcct gggcaag 1407

```

```

<210> SEQ ID NO 15
<211> LENGTH: 406
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: human a-2,6 sialic acid transferase

```

```

<400> SEQUENCE: 15

```

```

Met Ile His Thr Asn Leu Lys Lys Lys Phe Ser Cys Cys Val Leu Val
1             5             10             15

```

```

Phe Leu Leu Phe Ala Val Ile Cys Val Trp Lys Glu Lys Lys Lys Gly
20             25             30

```

```

Ser Tyr Tyr Asp Ser Phe Lys Leu Gln Thr Lys Glu Phe Gln Val Leu
35             40             45

```

```

Lys Ser Leu Gly Lys Leu Ala Met Gly Ser Asp Ser Gln Ser Val Ser
50             55             60

```

```

Ser Ser Ser Thr Gln Asp Pro His Arg Gly Arg Gln Thr Leu Gly Ser
65             70             75             80

```

```

Leu Arg Gly Leu Ala Lys Ala Lys Pro Glu Ala Ser Phe Gln Val Trp
85             90             95

```

```

Asn Lys Asp Ser Ser Ser Lys Asn Leu Ile Pro Arg Leu Gln Lys Ile
100            105            110

```

```

Trp Lys Asn Tyr Leu Ser Met Asn Lys Tyr Lys Val Ser Tyr Lys Gly
115            120            125

```

```

Pro Gly Pro Gly Ile Lys Phe Ser Ala Glu Ala Leu Arg Cys His Leu
130            135            140

```

```

Arg Asp His Val Asn Val Ser Met Val Glu Val Thr Asp Phe Pro Phe
145            150            155            160

```

```

Asn Thr Ser Glu Trp Glu Gly Tyr Leu Pro Lys Glu Ser Ile Arg Thr
165            170            175

```

```

Lys Ala Gly Pro Trp Gly Arg Cys Ala Val Val Ser Ser Ala Gly Ser
180            185            190

```

```

Leu Lys Ser Ser Gln Leu Gly Arg Glu Ile Asp Asp His Asp Ala Val
195            200            205

```

```

Leu Arg Phe Asn Gly Ala Pro Thr Ala Asn Phe Gln Gln Asp Val Gly
210            215            220

```

```

Thr Lys Thr Thr Ile Arg Leu Met Asn Ser Gln Leu Val Thr Thr Glu
225            230            235            240

```

```

Lys Arg Phe Leu Lys Asp Ser Leu Tyr Asn Glu Gly Ile Leu Ile Val
245            250            255

```

```

Trp Asp Pro Ser Val Tyr His Ser Asp Ile Pro Lys Trp Tyr Gln Asn
260            265            270

```

```

Pro Asp Tyr Asn Phe Phe Asn Asn Tyr Lys Thr Tyr Arg Lys Leu His

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275	280	285
Pro Asn Gln Pro Phe Tyr Ile Leu Lys Pro Gln Met Pro Trp Glu Leu		
290	295	300
Trp Asp Ile Leu Gln Glu Ile Ser Pro Glu Glu Ile Gln Pro Asn Pro		
305	310	315 320
Pro Ser Ser Gly Met Leu Gly Ile Ile Ile Met Met Thr Leu Cys Asp		
	325	330 335
Gln Val Asp Ile Tyr Glu Phe Leu Pro Ser Lys Arg Lys Thr Asp Val		
	340	345 350
Cys Tyr Tyr Tyr Gln Lys Phe Phe Asp Ser Ala Cys Thr Met Gly Ala		
	355	360 365
Tyr His Pro Leu Leu Tyr Glu Lys Asn Leu Val Lys His Leu Asn Gln		
	370	375 380
Gly Thr Asp Glu Asp Ile Tyr Leu Leu Gly Lys Ala Thr Leu Pro Gly		
	385	390 395 400
Phe Arg Thr Ile His Cys		
	405	

<210> SEQ ID NO 16

<211> LENGTH: 1218

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: nucleotides sequence of human a-2,6 sialic acid transferase

<400> SEQUENCE: 16

```

atgatccaca ccaacctgaa gaagaagttc agctgctgcg tgctggtggt cctgctgttc      60
gccgtgatct gcgtgtggaa ggagaagaag aaaggcagct actacgacag cttcaagctg      120
cagaccaagg agttccaggt gctgaagagc ctgggcaagc tggccatggg cagcgacagc      180
cagagcgtgt ccagctcctc caccagcat cccacagag gcagacagac cctgggcagc      240
ctgagaggcc tggccaaggc caagcccag gccagcttcc aggtgtggaa caaggacagc      300
agcagcaaga acctgatccc cagactgcag aagatctgga agaactacct gagcatgaac      360
aagtacaagg tgagctacaa aggacccgga cccggcatca agttcagcgc cgaggccctg      420
aggtgccacc tgagagacca cgtgaacgtg agcatggtgg aagtgaccga cttccccttc      480
aacaccagcg agtgggaagg ctacctgccc aaggagagca tcaggaccaa ggctggcccc      540
tggggcagat gcgcgtggtg gacgagcgtt ggcagcctga agagctccca gctgggcaga      600
gagatcgacg accacgatgc cgtgctgagg ttcaatggcg ctcccaccgc caacttcag      660
caggacgtgg gcaccaagac cacaatccgg ctgatgaaca gccagctggt gacaaccgag      720
aagcggttcc tgaaggacag cctgtacaac gagggcatcc tgatcgtgtg ggatcccagc      780
gtgtaccaca gcgacatccc caagtggtag cagaatcccg actacaactt cttcaacaac      840
tacaagacct atagaaagct gcaccccaac cagcccttct acatcctgaa gccccagatg      900
ccctgggagc tgtgggacat cctgcaggag atcagccctg aagagatcca gcccaccct      960
ccctccagcg gcatgctggg cattatcatc atgatgaccc tgtgcgacca ggtggacatc     1020
tacgagttcc tgcccagcaa gagaaagacc gacgtgtgct actactatca gaagttcttc     1080
gacagcgctt gcaccatggg cgcctaccac cccctgctgt acgagaagaa cctggtgaag     1140
cacctgaacc agggcaccga cgaggacatc tacctgctgg gcaaagccac cctgcccggc     1200
ttcagaacca tccactgc                                     1218

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<210> SEQ ID NO 17
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (12)
<223> OTHER INFORMATION: Xaa is Lys or Gly
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (13)
<223> OTHER INFORMATION: Xaa is Glu, Gly, or Ser
<220> FEATURE:
<223> OTHER INFORMATION: IgD hinge variant

<400> SEQUENCE: 17

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa Xaa Lys Glu Lys
1             5             10             15

Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro
                20             25             30

```

```

<210> SEQ ID NO 18
<211> LENGTH: 49
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (31)
<223> OTHER INFORMATION: Xaa is Lys or Gly
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (32)
<223> OTHER INFORMATION: Xaa is Glu, Gly, or Ser
<220> FEATURE:
<223> OTHER INFORMATION: IgD hinge variant

<400> SEQUENCE: 18

Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr Thr Ala Pro
1             5             10             15

Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa Xaa
                20             25             30

Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys
                35             40             45

Pro

```

```

<210> SEQ ID NO 19
<211> LENGTH: 30
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: IgD hinge

<400> SEQUENCE: 19

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Lys Glu Lys Glu Lys
1             5             10             15

Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro
                20             25             30

<210> SEQ ID NO 20
<211> LENGTH: 425
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: FceRIa ECD-hinge-Fc1

<400> SEQUENCE: 20

```

Val 1	Pro	Gln	Lys	Pro 5	Lys	Val	Ser	Leu	Asn 10	Pro	Pro	Trp	Asn	Arg 15	Ile
Phe	Lys	Gly	Glu 20	Asn	Val	Thr	Leu	Thr 25	Cys	Asn	Gly	Asn 30	Asn	Phe	Phe
Glu	Val	Ser	Ser	Thr	Lys	Trp	Phe 40	His	Asn	Gly	Ser	Leu 45	Ser	Glu	Glu
Thr	Asn	Ser	Ser	Leu	Asn	Ile 55	Val	Asn	Ala	Lys	Phe 60	Glu	Asp	Ser	Gly
Glu 65	Tyr	Lys	Cys	Gln	His 70	Gln	Gln	Val	Asn	Glu	Ser	Glu	Pro	Val	Tyr 80
Leu	Glu	Val	Phe	Ser 85	Asp	Trp	Leu	Leu	Leu 90	Gln	Ala	Ser	Ala	Glu 95	Val
Val	Met	Glu	Gly	Gln	Pro	Leu	Phe	Leu	Arg 105	Cys	His	Gly	Trp	Arg	Asn
Trp	Asp	Val	Tyr	Lys	Val	Ile	Tyr 120	Tyr	Lys	Asp	Gly	Glu 125	Ala	Leu	Lys
Tyr	Trp	Tyr	Glu	Asn	His 135	Asn	Ile	Ser	Ile	Thr	Asn 140	Ala	Thr	Val	Glu
Asp 145	Ser	Gly	Thr	Tyr	Tyr 150	Cys	Thr	Gly	Lys	Val 155	Trp	Gln	Leu	Asp	Tyr 160
Glu	Ser	Glu	Pro	Leu 165	Asn	Ile	Thr	Val	Ile 170	Lys	Ala	Pro	Arg	Glu 175	Lys
Tyr	Trp	Leu	Gln 180	Arg	Asn	Thr	Gly	Arg 185	Gly	Gly	Glu	Glu 190	Lys	Lys	Lys
Glu	Lys 195	Glu	Lys	Glu	Glu	Gln	Glu 200	Glu	Arg	Glu	Thr	Lys 205	Thr	Pro	Glu
Cys	Pro 210	Ser	His	Thr	Gln	Pro 215	Leu	Gly	Val	Phe	Leu 220	Phe	Pro	Pro	Lys
Pro 225	Lys	Asp	Thr	Leu	Met 230	Ile	Ser	Arg	Thr	Pro 235	Glu	Val	Thr	Cys	Val 240
Val	Val	Asp	Val	Ser 245	Gln	Glu	Asp	Pro	Glu 250	Val	Gln	Phe	Asn	Trp 255	Tyr
Val	Asp	Gly	Val 260	Glu	Val	His	Asn	Ala 265	Lys	Thr	Lys	Pro	Arg	Glu 270	Glu
Gln	Phe 275	Asn	Ser	Thr	Tyr	Arg	Val 280	Val	Ser	Val	Leu	Thr 285	Val	Leu	His
Gln	Asp 290	Trp	Leu	Asn	Gly	Lys 295	Glu	Tyr	Lys	Cys	Lys 300	Val	Ser	Asn	Lys
Gly 305	Leu	Pro	Ser	Ser	Ile 310	Glu	Lys	Thr	Ile	Ser 315	Lys	Ala	Lys	Gly	Gln 320
Pro	Arg	Glu	Pro	Gln 325	Val	Tyr	Thr	Leu	Pro 330	Pro	Ser	Gln	Glu	Glu 335	Met
Thr	Lys	Asn	Gln 340	Val	Ser	Leu	Thr	Cys 345	Leu	Val	Lys	Gly	Phe 350	Tyr	Pro
Ser	Asp	Ile 355	Ala	Val	Glu	Trp	Glu 360	Ser	Asn	Gly	Gln	Pro 365	Glu	Asn	Asn
Tyr	Lys 370	Thr	Thr	Pro	Pro	Val 375	Leu	Asp	Ser	Asp	Gly	Ser 380	Phe	Phe	Leu
Tyr 385	Ser	Arg	Leu	Thr	Val	Asp 390	Lys	Ser	Arg	Trp	Gln	Glu	Gly	Asn	Val 400
Phe	Ser	Cys	Ser	Val 405	Met	His	Glu	Ala	Leu 410	His	Asn	His	Tyr	Thr	Gln 415
Lys	Ser	Leu	Ser	Leu	Ser	Leu	Gly	Lys							

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420                               425

<210> SEQ ID NO 21
<211> LENGTH: 425
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: FceRIa ECD-hinge-Fc2

<400> SEQUENCE: 21

Val Pro Gln Lys Pro Lys Val Ser Leu Asn Pro Pro Trp Asn Arg Ile
1                               5                               10                               15
Phe Lys Gly Glu Asn Val Thr Leu Thr Cys Asn Gly Asn Asn Phe Phe
20                               25                               30
Glu Val Ser Ser Thr Lys Trp Phe His Asn Gly Ser Leu Ser Glu Glu
35                               40                               45
Thr Asn Ser Ser Leu Asn Ile Val Asn Ala Lys Phe Glu Asp Ser Gly
50                               55                               60
Glu Tyr Lys Cys Gln His Gln Gln Val Asn Glu Ser Glu Pro Val Tyr
65                               70                               75                               80
Leu Glu Val Phe Ser Asp Trp Leu Leu Leu Gln Ala Ser Ala Glu Val
85                               90                               95
Val Met Glu Gly Gln Pro Leu Phe Leu Arg Cys His Gly Trp Arg Asn
100                              105                              110
Trp Asp Val Tyr Lys Val Ile Tyr Tyr Lys Asp Gly Glu Ala Leu Lys
115                              120                              125
Tyr Trp Tyr Glu Asn His Asn Ile Ser Ile Thr Asn Ala Thr Val Glu
130                              135                              140
Asp Ser Gly Thr Tyr Tyr Cys Thr Gly Lys Val Trp Gln Leu Asp Tyr
145                              150                              155                              160
Glu Ser Glu Pro Leu Asn Ile Thr Val Ile Lys Ala Pro Arg Glu Lys
165                              170                              175
Tyr Trp Leu Gln Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Gly
180                              185                              190
Ser Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys Thr Pro Glu
195                              200                              205
Cys Pro Ser His Thr Gln Pro Leu Gly Val Phe Leu Phe Pro Pro Lys
210                              215                              220
Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val Thr Cys Val
225                              230                              235                              240
Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe Asn Trp Tyr
245                              250                              255
Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro Arg Glu Glu
260                              265                              270
Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr Val Leu His
275                              280                              285
Gln Asp Trp Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val Ser Asn Lys
290                              295                              300
Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala Lys Gly Gln
305                              310                              315                              320
Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln Glu Glu Met
325                              330                              335
Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly Phe Tyr Pro
340                              345                              350
Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro Glu Asn Asn

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355	360	365
Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser Phe Phe Leu		
370	375	380
Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu Gly Asn Val		
385	390	395
Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His Tyr Thr Gln		
405	410	415
Lys Ser Leu Ser Leu Ser Leu Gly Lys		
420	425	

<210> SEQ ID NO 22
 <211> LENGTH: 444
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: FceRIa ECD-hinge-Fc3

<400> SEQUENCE: 22

Val Pro Gln Lys Pro Lys Val Ser Leu Asn Pro Pro Trp Asn Arg Ile
1 5 10 15
Phe Lys Gly Glu Asn Val Thr Leu Thr Cys Asn Gly Asn Asn Phe Phe
20 25 30
Glu Val Ser Ser Thr Lys Trp Phe His Asn Gly Ser Leu Ser Glu Glu
35 40 45
Thr Asn Ser Ser Leu Asn Ile Val Asn Ala Lys Phe Glu Asp Ser Gly
50 55 60
Glu Tyr Lys Cys Gln His Gln Gln Val Asn Glu Ser Glu Pro Val Tyr
65 70 75 80
Leu Glu Val Phe Ser Asp Trp Leu Leu Leu Gln Ala Ser Ala Glu Val
85 90 95
Val Met Glu Gly Gln Pro Leu Phe Leu Arg Cys His Gly Trp Arg Asn
100 105 110
Trp Asp Val Tyr Lys Val Ile Tyr Tyr Lys Asp Gly Glu Ala Leu Lys
115 120 125
Tyr Trp Tyr Glu Asn His Asn Ile Ser Ile Thr Asn Ala Thr Val Glu
130 135 140
Asp Ser Gly Thr Tyr Tyr Cys Thr Gly Lys Val Trp Gln Leu Asp Tyr
145 150 155 160
Glu Ser Glu Pro Leu Asn Ile Thr Val Ile Lys Ala Pro Arg Glu Lys
165 170 175
Tyr Trp Leu Gln Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala
180 185 190
Thr Thr Ala Pro Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly Glu Glu
195 200 205
Lys Lys Gly Ser Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr Lys
210 215 220
Thr Pro Glu Cys Pro Ser His Thr Gln Pro Leu Gly Val Phe Leu Phe
225 230 235 240
Pro Pro Lys Pro Lys Asp Thr Leu Met Ile Ser Arg Thr Pro Glu Val
245 250 255
Thr Cys Val Val Val Asp Val Ser Gln Glu Asp Pro Glu Val Gln Phe
260 265 270
Asn Trp Tyr Val Asp Gly Val Glu Val His Asn Ala Lys Thr Lys Pro
275 280 285
Arg Glu Glu Gln Phe Asn Ser Thr Tyr Arg Val Val Ser Val Leu Thr

-continued

290	295	300
Val Leu His Gln Asp Trp	Leu Asn Gly Lys Glu Tyr Lys Cys Lys Val	
305	310	315 320
Ser Asn Lys Gly Leu Pro Ser Ser Ile Glu Lys Thr Ile Ser Lys Ala		
	325	330 335
Lys Gly Gln Pro Arg Glu Pro Gln Val Tyr Thr Leu Pro Pro Ser Gln		
	340	345 350
Glu Glu Met Thr Lys Asn Gln Val Ser Leu Thr Cys Leu Val Lys Gly		
	355	360 365
Phe Tyr Pro Ser Asp Ile Ala Val Glu Trp Glu Ser Asn Gly Gln Pro		
	370	375 380
Glu Asn Asn Tyr Lys Thr Thr Pro Pro Val Leu Asp Ser Asp Gly Ser		
	385	390 395 400
Phe Phe Leu Tyr Ser Arg Leu Thr Val Asp Lys Ser Arg Trp Gln Glu		
	405	410 415
Gly Asn Val Phe Ser Cys Ser Val Met His Glu Ala Leu His Asn His		
	420	425 430
Tyr Thr Gln Lys Ser Leu Ser Leu Ser Leu Gly Lys		
	435	440

The invention claimed is:

1. A polypeptide dimer comprising:

a first monomer comprising an extracellular domain of an alpha subunit of an IgE Fc receptor (FcεRIα-ECD),

a second monomer comprising an extracellular domain of an alpha subunit of an IgE Fc receptor (FcεRIα-ECD),

wherein the FcεRIα-ECD of the first and second monomers comprises the amino acid sequence of SEQ ID NO: 1,

wherein the first monomer and the second monomer each contain an Fc region, and

wherein the Fc region and the FcεRIα-ECD, in the first and second monomers, are linked via a hinge, and

wherein the polypeptide dimer comprises sialic acid and, a molar ratio of the sialic acid to the polypeptide dimer is 12 to 25.

2. The polypeptide dimer according to claim 1,

wherein the Fc region of the first and second monomers comprises the amino acid sequence of SEQ ID NO: 2.

3. The polypeptide dimer according to claim 1,

wherein the hinge of the first and second monomers is a hinge region derived from immunoglobulin IgD or a variant thereof.

4. The polypeptide dimer according to claim 3,

wherein the hinge region derived from immunoglobulin IgD or a variant thereof contains at least one cysteine.

5. The polypeptide dimer according to claim 3,

wherein the hinge region derived from immunoglobulin IgD or a variant thereof comprises the sequence of

(SEQ ID NO: 17)

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa1

Xaa2 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr

Lys Thr Pro Glu Cys Pro,

wherein Xaa1 is Lys or Gly, and

Xaa2 is Glu, Gly, or Ser; or

(SEQ ID NO: 18)

Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr

Thr Ala Pro Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly

Glu Glu Lys Lys Xaa3 Xaa4 Lys Glu Lys Glu Glu Gln

Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro,

wherein Xaa3 is Lys or Gly, and

Xaa4 is Glu, Gly, or Ser.

6. The polypeptide dimer according to claim 3,

wherein the hinge region derived from immunoglobulin IgD or a variant thereof comprises any one amino acid sequence selected from the group consisting of SEQ ID NO: 3, SEQ ID NO: 4, and SEQ ID NO: 19.

7. The polypeptide dimer according to claim 1,

wherein the sialic acid is N-acetylneuraminic acid.

8. A composition comprising a polypeptide dimer according to claim 1.

9. The composition according to claim 8, wherein the composition is for subcutaneous injection.

10. The composition according to claim 8, which is a pharmaceutical composition or a food composition.

11. The composition according to claim 8, wherein the composition is a pharmaceutical composition in a form of a transdermal patch or topical patch.

12. The composition according to claim 8,

wherein the Fc region of the first and second monomers comprises the amino acid sequence of SEQ ID NO: 2; and

wherein the hinge is a hinge region derived from immunoglobulin IgD or a variant thereof, and wherein the sialic acid is N-acetylneuraminic acid.

13. The composition according to claim 8,

wherein the hinge region derived from immunoglobulin IgD or a variant thereof comprises the sequence of

(SEQ ID NO: 17)

Arg Asn Thr Gly Arg Gly Gly Glu Glu Lys Lys Xaa1

Xaa2 Lys Glu Lys Glu Glu Gln Glu Glu Arg Glu Thr

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Lys Thr Pro Glu Cys Pro,

wherein Xaa1 is Lys or Gly, and
Xaa2 is Glu, Gly, or Ser; or

10

(SEQ ID NO: 18)

Ala Gln Pro Gln Ala Glu Gly Ser Leu Ala Lys Ala Thr

Thr Ala Pro Ala Thr Thr Arg Asn Thr Gly Arg Gly Gly

Glu Glu Lys Lys Xaa3 Xaa4 Lys Glu Lys Glu Glu Gln

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Glu Glu Arg Glu Thr Lys Thr Pro Glu Cys Pro,

wherein Xaa3 is Lys or Gly, and
Xaa4 is Glu, Gly, or Ser.

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